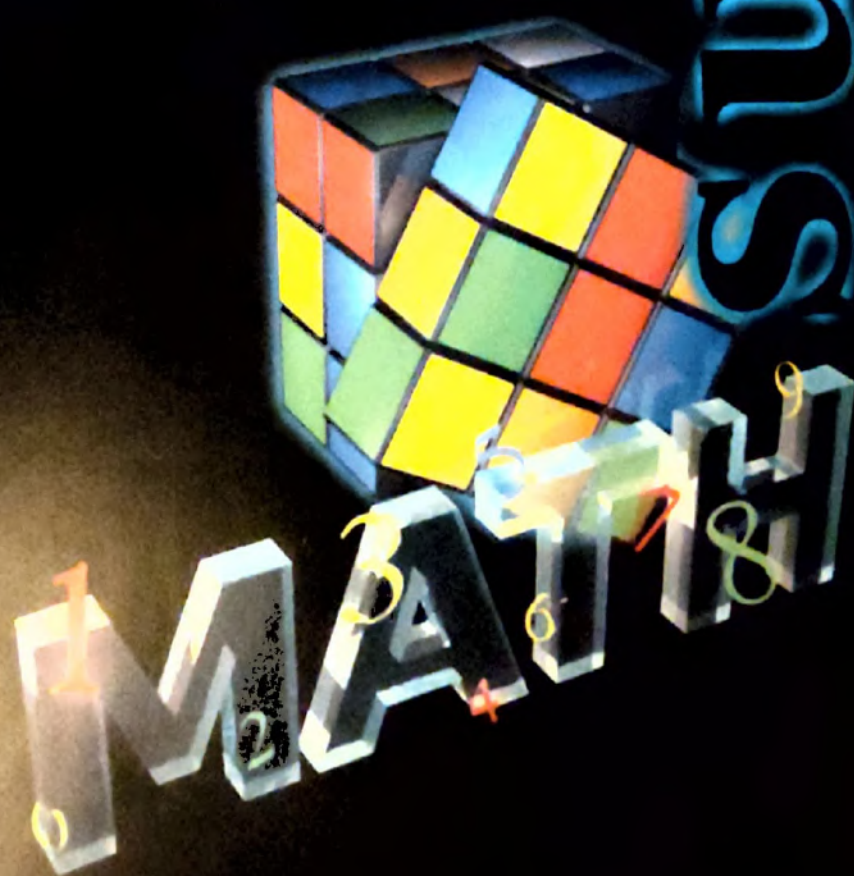




Dept. of Math | 2001
Stuyvesant H.S. | Vol 75, No.1

Math Survey

Volume LXXV, Number 1
Spring 2001

The Mathematics Publication of Stuyvesant High School



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From the Editors-in-Chief

This is the 75th publication of Math Survey, and we are extremely pleased with the finished product. Those of you who have read Math Survey before will notice a marked difference in the quality of the issue: it's finally in color!

Math Survey has changed in other ways as well (since last year and since its first publication). We've expanded the size of the issue. We use computers rather than VD1 kits to make diagrams. Most of our communication is done via email, and any student may join the staff.

We would like to thank the staff members and all the authors who made this anniversary edition of Math Survey possible. We would also like to acknowledge the math department faculty members for their invaluable help and direct contributions. Furthermore, we are thankful to the Student Union for its support of this publication.

We want you to know that we had a lot of fun putting this issue together. Many people worked to make this publication happen, and it's a tremendous source of pride and satisfaction to see months of planning and activity finally come to fruition.

Enjoy!

Dmitriy Aronov, Aileen Nielsen, Ming Jack Po

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THE ROTHENBERG MEMORIAL

Bruce Winokur

In the spring of 1997, Richard Rothenberg, the legendary Stuyvesant High School educator, chairman of the mathematics department, died suddenly at the age of 50. At that time, the teachers of the mathematics department, feeling that Dr. Rothenberg was worthy of memorializing, organized an effort to create a memorial that would speak of his devotion to mathematics and to the students and faculty of Stuyvesant. The results of that effort stand proudly on the fourth floor for the entire Stuyvesant family's benefit.

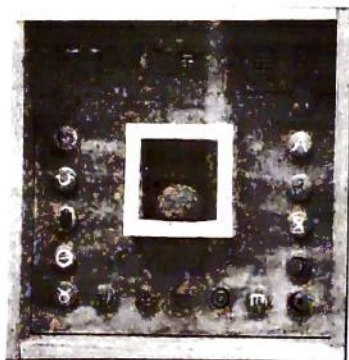
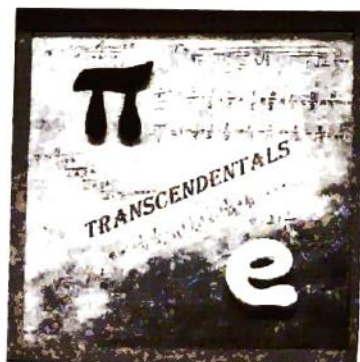
You will notice a plaque outside the math office honoring the recipients of our annual Richard Rothenberg Scholarship, awarded for highest honors in mathematics. We are proud to have the names of brother and sister Peter Graham and Kirsten Wickelgren, as well as that of Kamaldeep Ghandi on the wall, and we look forward to many more.

In addition, you can't help but notice the Richard Rothenberg Memorial, the centerpiece of the Richard Rothenberg Mathematics wing. This work, entitled "Celebration," was produced by Madeleine Segall-Marx, an artist and sculptor, and the mother of two Stuyvesant students. It celebrates both the mathematics that Dr. Rothenberg so dearly loved and passed on to so many Stuyvesantians, and the unique qualities that made him so special.

Behind a wall of glass, you will see a 7×7 matrix of cubical boxes with the 50th box embedded at the centroid of the piece. The 50 boxes are symbolic of Dr. Rothenberg's 50 years of life on this earth. As you peruse the boxes, you will see a collage of sculpture, painting, drawing, prose, and poetry, which lead you through both Dr. Rothenberg's world of mathematics and his contributions to his beloved students and colleagues.

Box₁₁ kicks off the memorial with the first counting number. Another number, 18, whose Hebrew symbol represents Life, constitutes a different box. The complex number i and the transcendental numbers π and e grace other boxes.

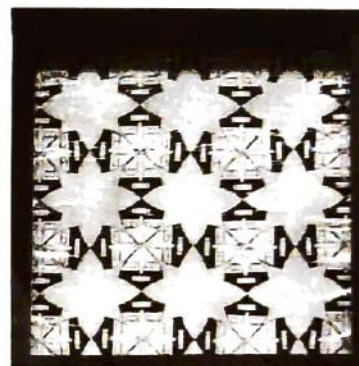
The embedded box, which may be considered the 50th, has a dual meaning, as do many others. The stone resting in the center of the box is symbolic of the Jewish tradition of leaving a stone on the grave of a loved one you've visited, who has moved on. In addition, the French word "calculus", means "stone," as humans first started calculating with little stones. This stone is surrounded by ancient symbols representing man's early attempts at quantification.



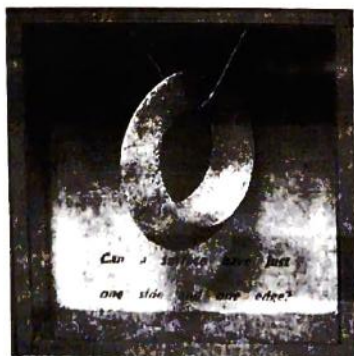
A number of other boxes serve directly as tributes to Dr. Rothenberg. Below the embedded box sits a heart of gold, described by an equation in three variables, representing the immense heart that was that of Dr. Rothenberg. Below that box, seemingly etched in stone, are excerpts from an open letter written to Dr. Rothenberg by a colleague after Richard's death and read at the memorial service in Stuyvesant's Murray Kahn Theatre.

Scattered throughout the piece are quotes from the writings of a number of Dr. Rothenberg's students, attesting to their love, affection, and respect for him, as well as his significant impact on their lives. In that spirit, another box contains a quote from the Talmud, "To be an ideal teacher, one must have an understanding heart and a sense of awe at the mystery of life and of other human beings. The mere act of teaching implies that one wishes the world well." In another box, there is a moving poem written by one whose friend and colleague had passed away. A naked winter tree is joined to the poem. Elsewhere, you will see a painting, into which one can read layers of meaning.

Some of the boxes refer to a few of the favorite problems of Dr. Rothenberg and his students, which can be solved using differential equations. There is his snow plow problem, his Battle of Trafalgar problem, and his Romeo and Juliet problem. The problems and their solutions, as depicted by students of Dr. Rothenberg's final differential equations classes, now hang in two-dimensional form in Room 405, across the hall from the memorial. Stuyvesant High School itself, to which Dr. Rothenberg devoted a good part of his life, is represented in one of the boxes by a tessellation of the plane using the Stuyvesant seal.



A few of the boxes refer to some important theorems of mathematics, such as the Pythagorean Theorem, the Fundamental Theorem of Calculus, Fermat's Last Theorem, and the Euler Relation. Accompanying the latter is a quote of the Harvard mathematician, Benjamin Pierce, "That is surely true, absolutely paradoxical; we cannot understand it and we don't know what it means, but we have proved it, and therefore we know it must be the truth."

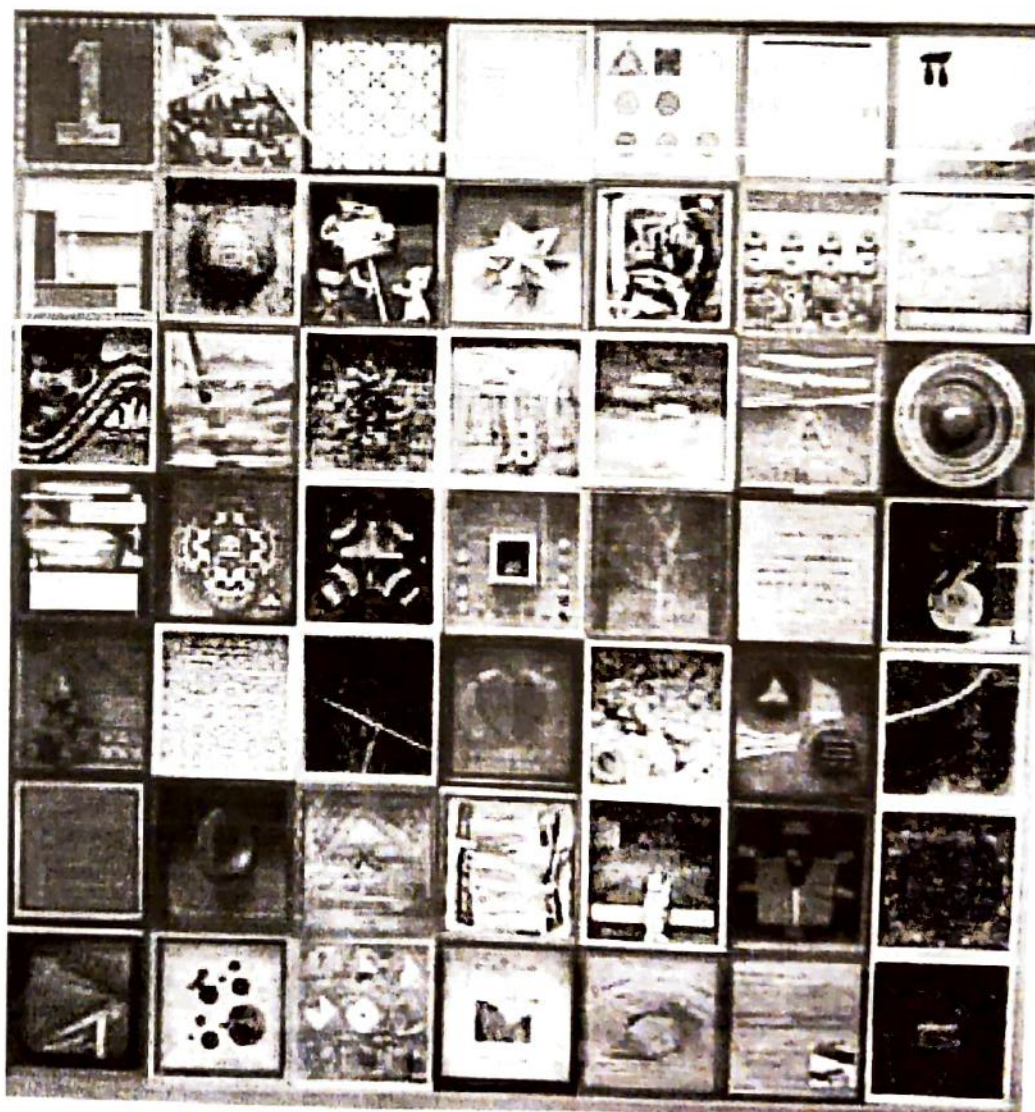


There are also boxes dealing with some well-known problems of mathematics: Apollonius' circle problem, the 16 point circle, the brachistocrone problem, and box-packing. Other boxes deal with such geometric figures as the parhexagon, the conic sections, the sphere, the polyhedra, and the Möebius string. Yet others involve topics such as probability, the cycloid, the quintic equation, Pascal's pyramid, non-Euclidean geometry, fractal geometry, chaos, and

transfinite numbers. Applied mathematics takes its place with boxes on information, DNA and knot theory, along with one concerning the inviolability of the speed of light.

You've read about some highlights of the Rothenberg Memorial. Even though in this case, the whole is greater than the sum of its parts, one owes it to oneself significant time spent on the individual components. The legacy of Richard Rothenberg lives on, at least in part through this memorial. Just as Dr. Rothenberg brought out the best in his students and colleagues when he was alive, through both our memories and this memorial, he continues to do so.

You'll often see a student or a group of students poring over a box, developing important insights and inspiration that will serve them well for the rest of their lives. Even though we are no longer graced with the physical presence of Richard Rothenberg, he will always be a part of us.



THE FOUR COLOR PROBLEM

Tonislav Ivanov

The Four Color Problem arose from the assertion of mapmakers that no more than four colors are necessary to color any map. A map is considered to be *colored* if no two countries (i.e. faces, being regions fully enclosed by curves) that meet along an edge have the same color. Non-adjacent countries and countries meeting at a vertex, however, are allowed to have the same color.

The terms used to describe a map – countries, edges, and vertices, are all illustrated below.

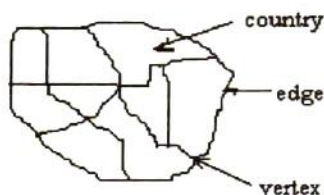


Fig. 1

The unbounded region (ocean) should also be differently colored from any of its adjacent countries. The shapes of the countries of a map are irrelevant to the coloring properties, as long as we know which countries share a border. Maps like this, which are essentially (mathematically speaking) the same are called *isomorphic*. Isomorphism suggests a very close relation between maps and graphs. A map can be defined as a planar graph (a finite collection of vertices and edges in the plane without edges crossing over) whose vertices have a valence (the valence of a vertex is the number of edges meeting at a that vertex, where loops are counted twice) of at least three, and whose faces have boundaries that are simple closed curves.

One powerful tool in proving the Four Color Conjecture is the Kempe chain. A *Kempe chain* is a chain of all countries on a map that are colored using two colors and linked together in a way that makes it possible to travel from one country to any other without going across a vertex. The countries in such a chain must be alternately colored (fig. 2). The main idea behind a Kempe chain is that interchanging the two colors in the chain will not affect the map's coloring property, that is, make a necessity for a new color.

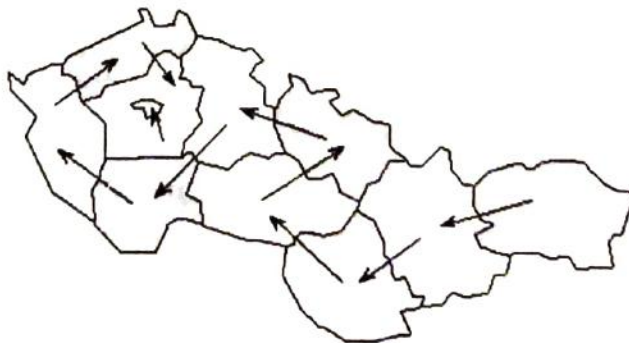


Fig. 2 Kempe chain on the map of former Czechoslovakia

A map may be composed of one or several connected regions. If each of the connected regions of a map can be four-colored, then the whole map can be four-colored. This can be easily verified through the use of Kempe chains. Suppose the coloring of different connected regions assigned different colors to the ocean. Then, we can pick one color and simply interchange colors in the chains of those regions where the ocean was colored differently. This way, the ocean will have same color throughout the map. Thus, the Four Color Problem is reduced to connected maps.

An important property of planar graphs is *duality*. To draw the dual of a planar graph G , place a dot on each face of the graph. These dots will be the vertices of the dual G^* . Whenever two faces of G meet at an edge, draw an arc joining the corresponding dots, so that it crosses the edge. This arc is an edge of G^* . If an edge belongs to only one face of G , draw a loop whose endpoints are both at the dot on the face. Clearly, the vertices of G correspond to faces of G^* , the faces of G correspond to vertices of G^* , and the edges of G correspond to the edges of G^* . The most important property of duality is that the dual of the dual of a connected planar graph is the original planar graph. Thus, coloring a map is the same as coloring the vertices of its dual.

A way of attacking the Four Color Conjecture is to ignore maps that are four-colorable and study only those that require at least five colors. Of course, it is now known that none exist, but while the conjecture was still unsolved, this method provided a promising route of attack. In fact, it was this method that led to the proof of the Four Color Conjecture. A map is *irreducible* if it requires at least five colors and all maps with fewer countries are four-colorable. Thus, the Four Color Conjecture is equivalent to the assertion that irreducible maps do not exist.

Suppose that a country with four or less edges occurred in an irreducible map. Patch this country, place a vertex on the patch, and connect the cut-off edges to that vertex. The map produced would be four-colorable, since it will have fewer countries than the original map. It can be arranged using Kempe chains, so that after removing the patch one of the four colors used will be available for the unveiled country, and thus the original map will be proved to be four-colorable. However, this contradicts the irreducibility of the original map, demonstrating that all countries of an irreducible map must have at least five edges.

It turns out to be easier to work with the dual graphs, instead of maps. Statement of the definition of an irreducible map in dual terms will be: An irreducible planar graph is one that requires at least five colors to color its vertices and has a minimum number of vertices. Irreducible graphs must be connected. If they were not, each of the connected parts, having fewer vertices than the minimum, would be four-colorable, leading to a four-colorability of the whole graph. It can be inferred that such graphs must have all of their vertices of valence five or greater. Irreducible graphs could be further cut down to graphs without loops and multiple edges, since removing such elements does not have an effect on the coloring properties of the graph. An important observation made about the Four Color Conjecture was that it is true if and only if it is true for all maps in which all vertices are 3-valent. Thus, irreducible planar graphs can be reduced to triangular graphs (where each face has three sides).

A collection of graphs with the property that any triangular graph must contain at least one graph of the collection is called an unavoidable set. Each member of an unavoidable set of graphs would be called a configuration. A reducible configuration would be any graph that cannot exist as a subgraph of a triangular irreducible graph. Once the existence of an unavoidable set of reducible configurations is shown, the Four Color Theorem is proven. Appel and Haken used what is called a discharging method to find their unavoidable set. This process involves assigning an integer to each vertex, thought of as an electrical charge, and redistributing the

charges according to certain rules, called discharging rules. Different discharging rules can lead to different unavoidable sets. Through a process of trial and error, they arrived at a discharging method that produced an unavoidable set that they believed consisted of reducible configurations. The unavoidable set consisted of 1,936 graphs, all generated by hand. These graphs were tested for reducibility by a computer and according to the computer, all of the graphs in the unavoidable set were reducible.

One interesting application the Four Color Theorem is its use to possibly explain the phenomenon of human color vision. This intricate human capability is not deeply understood and there is no clear explanation of how color vision functions. Nonetheless, it is apparent that color vision has evolved to enable us to optimally distinguish between objects. Despite the three-dimensional experience, the scenes we see are derived from two-dimensional image focused on the retina of the eye. This fact establishes the relationship between color vision and the Four Color Problem. Unlike maps, however, the visual system cannot arbitrarily assign colors to different regions to avoid ambiguity. On the contrary, it must deal with the spectral returns (the different wavelengths of the visible light spectrum reflected by different objects) whatever they happen to be. It was proposed by the German physiologist Ewald Hering that all colors that we experience are perceptual mixtures of four primary hues – red, green, blue, and yellow – and the visual system differentiates between objects by performing comparisons of two pairs of opponents – blue-yellow and red-green. As a consequence of the comparisons in these four different dimensions (red, green, blue, yellow), the spectral return from any given region is experienced as relatively more reddish, greenish, bluish, or yellowish than the return from any other region in the scene. Colorblind individuals are deficient either in distinguishing blues and yellows, or more commonly reds and greens. Such individuals are capable of carrying out only two of the necessary four color comparisons, and are thereby at a disadvantage in distinguishing objects on the bases of spectral differences. Fewer than four dimensions of comparison will not suffice. However, it is suggested that the sufficiency of four dimensions has arisen as a natural solution of the Four Color Problem by the visual system.

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JONES POLYNOMIAL IN KNOT THEORY

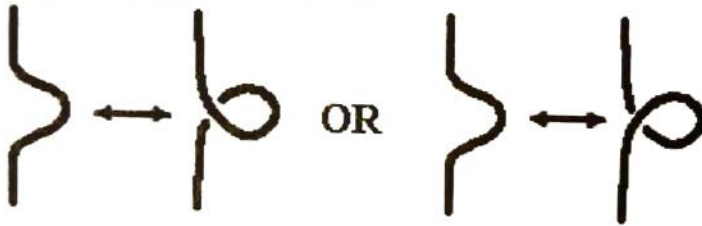
Artem Ivashchenko

The central topic of knot theory is knot equivalence: how can we prove (or disprove) that two knots are equivalent? A *knot* K is a twisted closed loop in 3D. A collection of knots, which may or may not be connected to each other in a chain-like manner, is called a *link* L . One knot all by itself is a trivial link of one component, so everything said about knots also holds true for links.

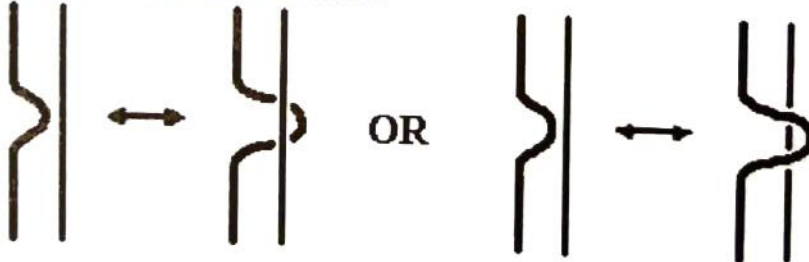
A projection of a knot onto a plane may look something like this (this one is called a trefoil knot), and in 1924 a German mathematician Kurt Reidemeister proved that two knot projections are related to one another by any sequence of three moves shown below if and only if the two projections are of the same knots. Hence, the two knots are equivalent.



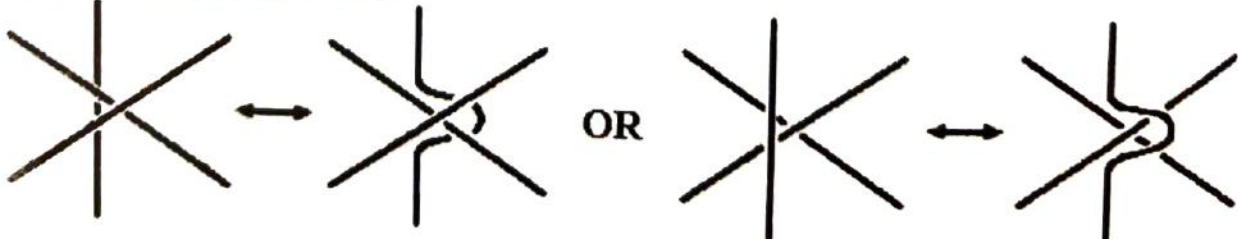
Type I Reidemeister Move



Type II Reidemeister Move

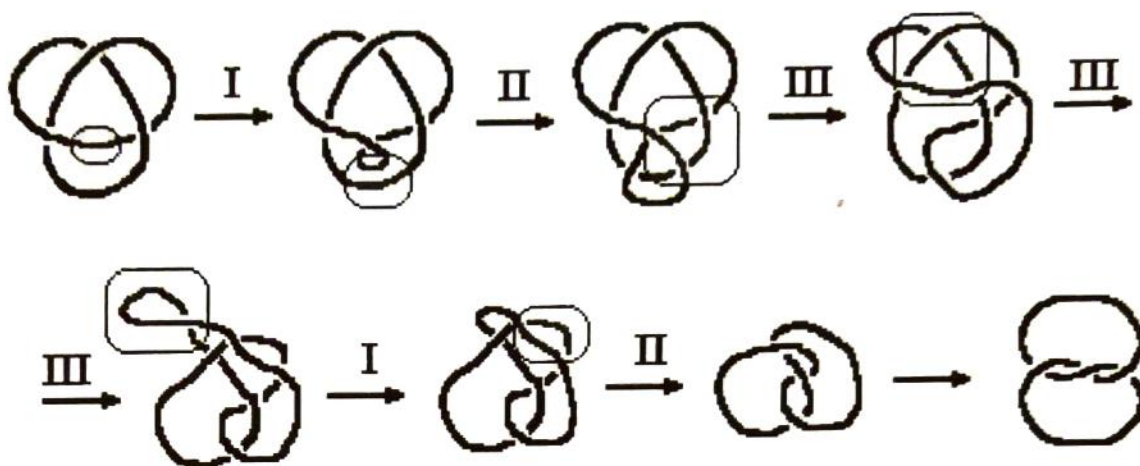


Type III Reidemeister Move



However, that procedure can be quite lengthy and inefficient. For example, to prove that these two projections of the simplest non-trivial knot are actually equivalent, we need the following sequence.





So what we need is a simpler, more computer-friendly way to show the equivalence of two knots. Here is where Jones polynomial of a knot comes into play.

Jones polynomial was discovered in 1984 by Vaughan Jones, a mathematician from New Zealand. The polynomial is computed from a knot projection, but, being a knot invariant, it is the same for different projections of a knot. Therefore, if two knots projections have different Jones polynomials, they represent different knots. We will now derive Jones polynomial and prove that it indeed is a knot invariant.

The approach used here to construct a Jones polynomial is through a bracket polynomial (denoted $\langle K \rangle$), developed by Louis Kauffman. The first rule for constructing a bracket polynomial is: a polynomial of an unknot (a trivial knot with no crossings, i.e. circle) is 1. That is done simply for convenience.

$$\text{Rule 1: } \langle \bigcirc \rangle = 1$$

Second, we need a method for finding a bracket polynomial of a knot (or link) in terms of simpler knots (or links). To do that, we use one of the skein relations. Skein relations are equations that correlate a polynomial of a link to polynomials of links obtained by changing the crossing in the projection of the original link. The skein relation used for bracket polynomial is as follows (A and B are coefficients that we still have to figure out):

$$\begin{aligned} \text{Rule 2: } \langle \text{X} \rangle &= A \langle \text{>} \rangle + B \langle \text{<} \rangle \\ \langle \text{X} \rangle &= A \langle \text{<} \rangle + B \langle \text{>} \rangle \end{aligned}$$

The second equation is simply the first one rotated 90 degrees.

Finally, we need a rule for adding an unknot to a link, the result of which will always be a disconnected, or split, link.

$$\text{Rule 3: } \langle L \cup \bigcirc \rangle = C \langle L \rangle$$

Again, in this case C is just a coefficient.

Since the bracket polynomial needs to be a knot invariant, we must see that it is not changed under the three Reidemeister moves. The computations for the Type II move are as follows:

$$\begin{aligned}
\langle \text{X} \rangle &= A \langle \text{Y} \rangle + B \langle \text{Z} \rangle \\
&= A(A \langle \text{X} \rangle + B \langle \text{Y} \rangle) + B(A \langle \text{Z} \rangle + B \langle \text{X} \rangle) \\
&= A(A \langle \text{X} \rangle + BC \langle \text{Y} \rangle) + B(A \langle \text{Z} \rangle + B \langle \text{X} \rangle) \\
&= (A^2 + ABC + B^2) \langle \text{Y} \rangle + BA \langle \text{Z} \rangle \stackrel{?}{=} \langle \text{Y} \rangle
\end{aligned}$$

Just like in any other expression, to ensure equality, we need to make coefficients for the same variables equal on both sides of our expression. Hence: $AB=1$, or $B=A^{-1}$; $A^2 + C + A^{-2} = 0$, or $C = -A^2 - A^{-2}$.

In light of these new formulas, we can rewrite the three rules for bracket polynomial as follows:

Rule 1: $\langle \bigcirc \rangle = 1$

Rule 2: $\langle \text{X} \rangle = A \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle$
 $\langle \text{X} \rangle = A \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle$

Rule 3: $\langle L \cup \bigcirc \rangle = (-A^2 - A^{-2}) \langle L \rangle$

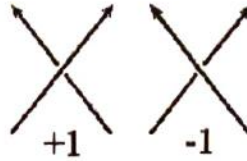
Under these rules, the Type III Reidemeister move also holds:

$$\begin{aligned}
\langle \text{X} \rangle &= A \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle = \\
&= A \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle = \langle \text{X} \rangle
\end{aligned}$$

And finally, we need to check for the Type I Reidemeister move:

$$\begin{aligned}
\langle \text{X} \rangle &= A \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle \\
&= A(-A^2 - A^{-2}) \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle \\
&= -A^3 \langle \text{Y} \rangle \\
\langle \text{X} \rangle &= A \langle \text{Y} \rangle + A^{-1} \langle \text{Z} \rangle \\
&= A \langle \text{Y} \rangle + A^{-1}(-A^2 - A^{-2}) \langle \text{Y} \rangle \\
&= -A^{-3} \langle \text{Y} \rangle
\end{aligned}$$

This, however, does not work out to our advantage. The two expressions must be the same, yet they differ by the sign of their power. To correct that problem we introduce an orientation to a knot, i.e. we give a direction to the string it is composed of. Then, for every projection we assign a +1 or a -1 in accordance with this rule:



The sum of all ± 1 for a knot (link) is called a writhe or a twist of a knot (link) and is denoted $\omega(\mathbf{K})$ or $\omega(\mathbf{L})$. Hence, for our oriented trefoil knot shown here, $\omega(\mathbf{K}) = -3$.



We can verify that $\omega(\mathbf{L})$ is an invariant for Type II and Type III move (it stays 0 for Type II and $\pm 1, \pm 3$ for Type III), but changes by ± 1 for Type I move. So we define a new polynomial $X(\mathbf{L})$ as $X(\mathbf{L}) = (-A^3)^{-\omega(\mathbf{L})} \langle \mathbf{L} \rangle$.

Suppose we take a loop out of the following link. Then we can show that $X(\mathbf{L}) = X(\mathbf{L}')$, hence making $X(\mathbf{L})$ a knot invariant.

L'

$X(\mathbf{L}) = x - 1$

$\Rightarrow$

L

$X(\mathbf{L}') = x$

$$\begin{aligned}
 X(\mathbf{L}') &= (-A^3)^{-\omega(\mathbf{L}')} \langle \mathbf{L}' \rangle \\
 &= (-A^3)^{-\omega(\mathbf{L})-1} \langle \mathbf{L}' \rangle \\
 &= (-A^3)^{-\omega(\mathbf{L})-1} (-A^3) \langle \mathbf{L} \rangle \\
 &= (-A^3)^{-\omega(\mathbf{L})} \langle \mathbf{L} \rangle = X(\mathbf{L})
 \end{aligned}$$

$X(\mathbf{L}') = X(\mathbf{L}) - 1$

Note: we know that $\langle \mathbf{L}' \rangle = (-A^3) \langle \mathbf{L} \rangle$ from our earlier computation of bracket polynomial for Type I Reidemeister move.

We have finally derived a polynomial invariant for knots. Using this polynomial, we can verify what our earlier sequence of Reidemeister moves showed us.

$$X(\bigcirc) = X(\bigcirc) = -A^{16} + A^{12} + A^4$$

The original Jones polynomial is obtained from $X(\mathbf{L})$ by substituting each A with $t^{1/4}$, so the Jones polynomial for any projection of the trefoil knot would be $-t^4 + t^3 + t$.

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THE ELUSIVE LINK BETWEEN π AND THE P SERIES

Antony Chow

"If anyone finds and communicates to us that which thus far has eluded our efforts, great will be our gratitude..."

This famous statement was made in the town of Basel by Jakob Bernoulli, in his mathematical work *Tractatus*. Though elegant, this statement was nothing more than a public concession of defeat to an intractable problem – the solution of which would prove a difficult task even for a mathematician who was acknowledged as one of the premier mathematicians of Europe in the 17th century. Bernoulli's failure to solve this problem is by no means ignominious, for this problem had likewise stumped the rest of the mathematical community of the time. This problem finally met its match in the 18th century, in the hands of a 24-year-old mathematician, whose answer seemed to foreshadow his rise in the ranks of the greatest mathematicians of all time. This mathematician was Leonhard Euler.

Well, enough with the suspense. What exactly is this notorious, Basel problem? It is the computation of an infinite series of the form

$$1 + \frac{1}{2^P} + \frac{1}{3^P} + \frac{1}{4^P} + \frac{1}{5^P} + \frac{1}{6^P} + \dots$$

where the exponent P is a positive integer greater than or equal to 2, since the series diverges when P is equal to 1. The reason for the series's convergence to a finite sum when $P \geq 2$ is simple. When we observe another infinite series, we see that

$$\begin{aligned} 1 + \frac{1}{3} + \frac{1}{6} + \frac{1}{10} + \frac{1}{15} + \frac{1}{21} + \dots &= 2 \cdot \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{12} + \frac{1}{20} + \frac{1}{30} + \frac{1}{42} + \dots \right) \\ &= 2 \cdot \left(1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \frac{1}{3} - \frac{1}{4} + \frac{1}{4} - \frac{1}{5} + \dots \right) = 2 \cdot 1 = 2 \end{aligned}$$

Because we know that $\frac{1}{k^2} < \frac{1}{k(k+1)/2}$ for all $k \geq 2$, we conclude that

$$\begin{aligned} 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \dots &< 1 + \frac{1}{3} + \frac{1}{6} + \frac{1}{10} + \frac{1}{15} + \frac{1}{21} + \dots \\ 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \frac{1}{5^2} + \frac{1}{6^2} + \dots &< 2 \end{aligned}$$

Thus, the P series for $P=2$ converges to a finite sum less than 2. Since all P series for $P > 2$ are smaller than the above series, we can conclude that all P series for $P \geq 2$ will converge to a finite value.

Although mathematicians know that the P series for $P=2$ converges to a finite sum, they still can't calculate the sum. Euler showed that this series converges to $\pi^2/6$, a sum that relates the number associated with circles to the sum of square numbers. Even more remarkable, Euler did so using a well-known trigonometric equation that every good calculus student knows:

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \dots$$

We can create a function $f(x)$ such that

$$* f(x) = \frac{\sin x}{x} = 1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots, \text{ where } x \neq 0$$

We also define $f(0)=1$ and denote the above as equation *. Note that the roots of $f(x)$ can be obtained by solving $\sin x = 0$, since all roots occur at $x \neq 0$. Thus, the roots are

$$x = \pm\pi, \pm 2\pi, \pm 3\pi, \pm 4\pi, \dots$$

Hence, we have defined a function $f(x)$, where $f(0)=1$ and all roots are known. If $P(x)$ is a polynomial of degree n with n roots, $a, b, c, \dots, d$, and $P(0)=1$, then we can determine what the polynomial is. The polynomial will clearly be

$$P(x) = (1 - x/a) \cdot (1 - x/b) \cdot (1 - x/c) \cdot \dots \cdot (1 - x/d)$$

Knowing this about polynomials, let's turn our attention back to $f(x)$, where we knew that $f(0)=1$, and the roots are $x = \pm\pi, \pm 2\pi, \pm 3\pi, \pm 4\pi, \dots$. As we can see, there is an infinite number of roots of $f(x)$. However, Euler made a bold move and attempted to write the equation for $f(x)$ using the same logic as above!

$$\begin{aligned} f(x) &= \left(1 - \frac{x}{\pi}\right) \cdot \left(1 + \frac{x}{\pi}\right) \cdot \left(1 - \frac{x}{2\pi}\right) \cdot \left(1 + \frac{x}{2\pi}\right) \cdot \left(1 - \frac{x}{3\pi}\right) \cdot \left(1 + \frac{x}{3\pi}\right) \cdot \dots \\ &= \left(1 - \frac{x^2}{\pi^2}\right) \cdot \left(1 - \frac{x^2}{4\pi^2}\right) \cdot \left(1 - \frac{x^2}{9\pi^2}\right) \cdot \left(1 - \frac{x^2}{16\pi^2}\right) \cdot \dots \end{aligned}$$

Remembering equation *, we see that, remarkably, $f(x)$ is equal to an infinite sum *and* an infinite product! Equating the two values yields

$$1 - \frac{x^2}{3!} + \frac{x^4}{5!} - \frac{x^6}{7!} + \dots = \left(1 - \frac{x^2}{\pi^2}\right) \cdot \left(1 - \frac{x^2}{4\pi^2}\right) \cdot \left(1 - \frac{x^2}{9\pi^2}\right) \cdot \left(1 - \frac{x^2}{16\pi^2}\right) \cdot \dots$$

Euler imagined multiplying out the infinite product on the right and matching terms with equal powers of x . The term without an x would be 1. This corresponds to the 1 on the left side. Next, we collect all terms with x^2 . The only way to obtain such a term on the right is to multiply a 1 with an x^2 term, leaving it unchanged. Hence, after expanding the infinite product, we obtain

$$1 - \frac{x^2}{3!} + \dots = 1 - \left[\frac{1}{\pi^2} + \frac{1}{4\pi^2} + \frac{1}{9\pi^2} + \frac{1}{16\pi^2} + \dots \right] \cdot x^2 + \dots$$

Note that we don't need to know anything about terms with higher powers of x . What we have is enough. We have two equal polynomials, which both start with 1 and have a term with x^2 . We know that the coefficients of x^2 on both sides must be equal. Hence, we have the following:

$$\begin{aligned} -\frac{1}{3!} &= -\left(\frac{1}{\pi^2} + \frac{1}{4\pi^2} + \frac{1}{9\pi^2} + \frac{1}{16\pi^2} + \dots \right) \\ \frac{1}{6} &= \frac{1}{\pi^2} \left(1 + \frac{1}{4} + \frac{1}{9} + \frac{1}{16} + \dots \right) \\ \frac{\pi^2}{6} &= 1 + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{4^2} + \dots \end{aligned}$$

Here we have it – the amazing result. What makes this approach so remarkable is its elegance through simplicity. Euler’s brilliant insight that let these ordinary concepts play such a big role in solving the problem.

Euler was not content to stop merely at the evaluation of the P series at $P = 2$. He then successfully moved on to find an efficient way of evaluating the P series whenever P is any even integer. However, to this day, the P series for $P = 3, 5, 7, 9, \dots$ is undetermined. Perhaps π will be somehow involved in these answers too. One thing is certain – we have yet to fully explore the implications of the elusive link π has to this 300-year-old puzzle.

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TESSELLATIONS

Daniella Loh

Tessellations are geometric patterns or tilings. They are designs composed of repeating patterns of distinct shapes, and can be broken down into identical sections, which repeat throughout the design without gaps or overlaps on the plane. These shapes are typically polygons, although the term, tessellation, can apply to any type of closed figure.

Tessellations can be found in the most common places such as kitchen floors, mosaics, and stained glass windows. Practical in covering large surfaces with patterns, they have been used as an art form for thousands of years all over the world in a variety of cultures and forms. Twentieth century artist M.C. Escher was noted for creating tessellations by altering geometric tessellating shapes into forms such as birds, reptiles, fish, and people:



Fig. 1

In addition to artistic purposes, tessellations play an important role in science. They are used in x-ray crystallography, a field of science dealing with the natural repeating arrangements of ions in crystals. They are also linked to quantum mechanics, the study of subatomic particles, as well as a number of other fields in science and engineering.

The simplest case of a tessellation uses only one type of polygon, which is regular. These tessellations are called the regular tessellations. For example, squares, equilateral triangles, and regular hexagons can tessellate:

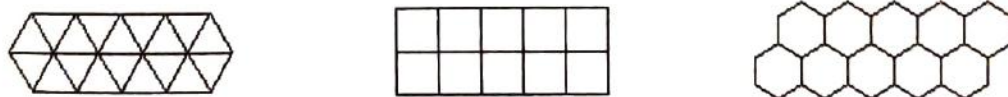


Fig. 2

These are the only three regular tessellations that exist.

Semi-regular tessellations also use regular polygons in patterning, but more than one polygon is used. The polygon arrangement at all vertices must be the same. In order to find tessellations such as these, it is first necessary to find a combination of regular polygons that will fill the node of 360° . If this pattern can be repeated to cover a plane, a tessellation exists.

There are 21 different types of regular polygon arrangements that can fill a node. However, only 8 of these, other than the included three regular tessellations (of the equilateral triangle, square, and hexagon), will actually tessellate. This can be found by trial and error. Here is a tessellation of one arrangement that does tessellate. Notice that the same arrangements are found at each vertex.

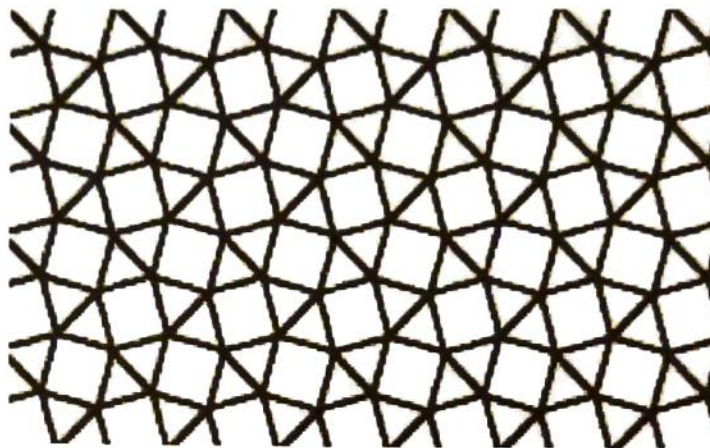


Fig. 3

In the previous type of tessellations the same polyhedra are found in the same order at every vertex. Demiregular tessellations are defined as tessellations of regular polygons in which there are either two or three different polygon arrangements. In other words, there is more than one type of a node. These tessellations are usually found using trial and error as well. There are at least 14 demiregular tessellations. More complex types of tessellations with more than one node (nonhomogeneous tessellations) may have more than three different polygon arrangements. In this case, there is an infinite number of possibilities.

Many modifications that can be applied to tessellation themes. Some of them allow looser rules, such as joining all elements edge to edge and corner to corner. Another variation involves squashing, stretching, and rotating the elements of a regular tessellation in a systematic way. Many figures, such as the hexagon, can be distorted in different ways without losing their ability to tessellate the plane. When lengths of edges are changed, interesting variations can be made as well.

There are also more advanced methods of tilings that are not necessarily tessellations. Rather, they are advanced variations of them. One of these is dissection tiling, which uses polygons in tiling without gaps or overlaps, just as the ones described previously. However, instead of repeating over a plane, they are split up into smaller polygonal pieces, or dissected, and these pieces are then rearranged periodically to create a standard tiling. The frequency of each of these smaller pieces, or how often they appear in the tiling, must be proportional to their area.

Another method is penrose tiling, in which a plane is tiled without repeating a pattern of shapes. The following is an example of this. If you look closely, there is repeating pattern.

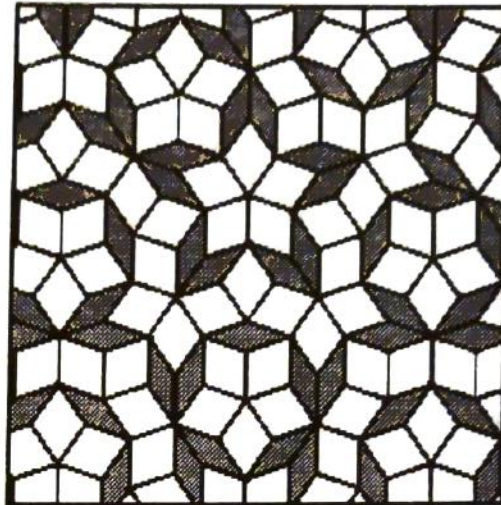


Fig. 4

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THREE-CORNERED DUEL

Johnny Wong

A, B, C are to fight a three-cornered pistol duel. All know A's chance of hitting a target is 0.3, C's is 0.5, and B never misses. They are to fire at their choice of target in succession in the order A, B, C, cyclically (but a hit man loses further turns and is no longer shot at) until only one man is left unhit. What should A's strategy be?

First, let's look at the option of shooting at C. If A shoots at C and hits him, then B will shoot at A during the next shot because A will be the only opponent left. Since B never misses, A will be definitely shot. This means there is no point in shooting at C because if A hits C, he will lose.

The answer may now seem like shooting at B but let's keep going. If A hits B, then A and C will be shooting at each other until one of them hits the other, with C shooting first. In the first round, A's chance of winning is 0.5 (C missing) $\times 0.3$ (A hitting). A's chance of winning in the second round is 0.5×0.5 (C missing in the 1st and 2nd round) $\times 0.7$ (A missing in the 1st round) $\times 0.3$ (A hitting in the 2nd round). We can continue this pattern to find the probability of A winning any given round. A's chance of winning one of the rounds is then the sum of the individual probabilities, which form the following geometric series:

$$(0.5)(0.3) + (0.5)^2(0.7)(0.3) + (0.5)^3(0.7)^2(0.3) + (0.5)^4(0.7)^3(0.3) + \dots$$

Using the formula for the sum of a geometric sequence we find the probability to be

$$\frac{(0.3)(0.5)}{1 - (0.5)(0.7)} = \frac{0.15}{0.65} = \frac{3}{13}$$

So, we found that A's chance of winning is $3/13$ if he hits B and 0 if he hits C. But wait! If A shoots either one of the opponents and *misses*, then B will shoot and hit C during the next shot because C is a more dangerous opponent (we are assuming that everyone thinks rationally). Then, A will have one shot at B with a $3/10$ chance hitting and winning. Since $3/10 > 3/13$, A's chances are better if he misses his first shot! This means that A's best strategy would be to *shoot into the ground* (forfeiting his 1st turn).

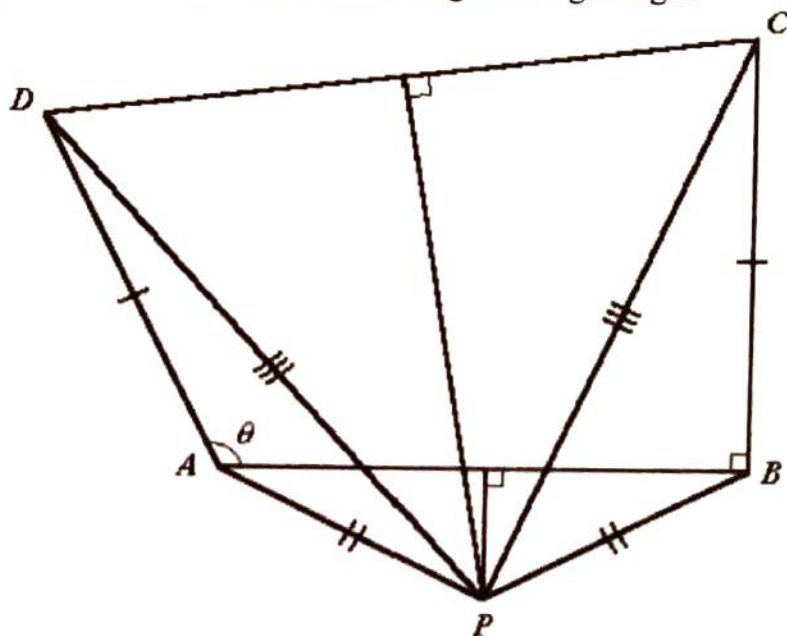
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THE FAULTY PROOF

Valerie Gorodetskaya and Michael Schnall-Levin

According to the proof below, any obtuse angle is a right angle!



Given an obtuse angle θ , we construct a quadrilateral $ABCD$, such that $\angle DAB = \theta$, $\angle ABC = 90^\circ$, and $AD = BC$. The perpendicular bisector of AB meets the perpendicular bisector of BC at some point P . It follows that $AP = BP$ and $CP = DP$. Then $\triangle ADP$ and $\triangle BCP$ have equal sides and are congruent. Thus $\angle DAP = \angle CBP$. Also, since $\triangle ABP$ is isosceles, $\angle BAP = \angle ABP$. Subtracting gives $\theta = \angle DAP - \angle BAP = \angle CBP - \angle ABP = 90^\circ$. Since θ was given as an arbitrary obtuse angle, we conclude that any obtuse angle is indeed a right angle.

This is obviously an absurd conclusion – there just *has* to be something wrong with the “proof”. Can you find the mistake?

For explanation see next page.

The figure used for the above proof is impossible to construct. In reality, the point of intersection P is much lower than what is shown, and the line segment DP is outside of AD . As a result, the real figure is different (fig. 1), and unreasonable conclusions cannot be made about θ .

Fig. 1

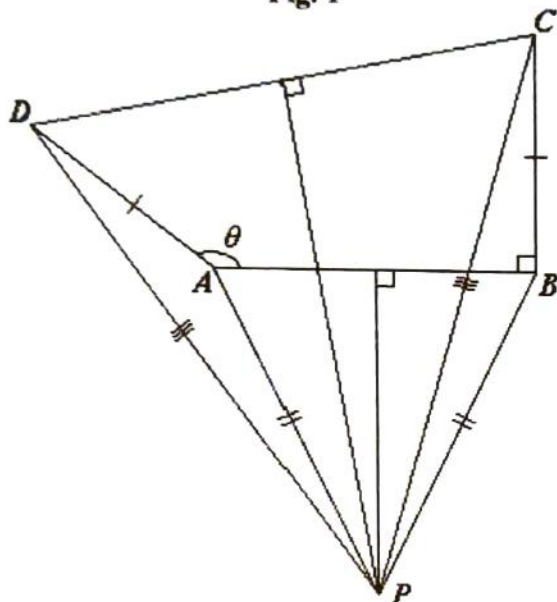
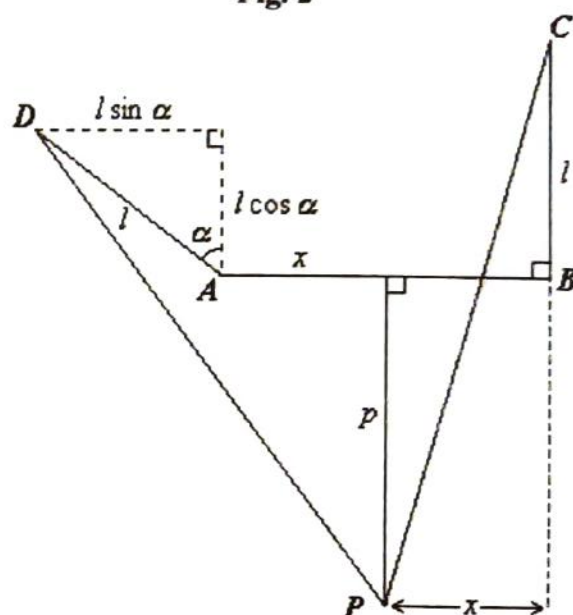


Fig. 2



To prove that line DP is always outside of the line AD , it can be shown that the slope of DP is always more negative than the slope of AD .

Construction (fig. 2):

1. A coordinate system is created. Point A is set to be $(0,0)$, and line segment AB is placed on the positive x -axis. The y -axis extends from point A perpendicular to line segment AB .
2. Angle α is the angle between line AD and the positive y -axis.
3. For simplicity, lengths of line segments AD and BC are labeled l .
4. The distance from point A to the perpendicular bisector of line segment AB is labeled x .

First, we find the y -coordinate of point P . Then we can find the slopes of lines DP and AD and compare them. We can find the lengths of line segments DP and CP using the Pythagorean Theorem. Since these lengths are equal, we can set up the following equation, and then solve it for p :

$$\sqrt{(l \sin \alpha + x)^2 + (l \cos \alpha + p)^2} = \sqrt{x^2 + (l + p)^2}$$

After squaring both sides and expanding, the equation becomes

$$l^2 \sin^2 \alpha + 2lx \sin \alpha + x^2 + l^2 \cos^2 \alpha + 2lp \cos \alpha + p^2 = x^2 + l^2 + 2lp + p^2$$

After canceling out like terms and applying the identity $\sin^2 \alpha + \cos^2 \alpha = 1$, we get

$$2lx \sin \alpha + 2lp \cos \alpha = 2lp$$

Solving for p ,

$$p = \frac{x \sin \alpha}{1 - \cos \alpha}$$

Note that the above expression represents the magnitude of p . Since point P lies below the x -axis, its y -coordinate is actually negative. We can now find the slope of DP :

$$m_{DP} = \frac{l \cos \alpha + \frac{x \sin \alpha}{1 - \cos \alpha}}{-l \sin \alpha - x}$$

To simplify this equation, we multiply the fraction at the top by $1 + \cos \alpha$, obtaining

$$\frac{x \sin \alpha}{1 - \cos \alpha} \cdot \frac{1 + \cos \alpha}{1 + \cos \alpha} = \frac{x \sin \alpha (1 + \cos \alpha)}{1 - \cos^2 \alpha} = \frac{x(1 + \cos \alpha)}{\sin \alpha}$$

Substituting this result into the original fraction and simplifying,

$$\begin{aligned} m_{DP} &= \frac{l \cos \alpha + \frac{x(1 + \cos \alpha)}{\sin \alpha}}{-l \sin \alpha - x} \\ &= \frac{l \cos \alpha \sin \alpha + x + x \cos \alpha}{(-1)(l \sin \alpha + x) \sin \alpha} \\ &= \frac{(l \sin \alpha + x) \cos \alpha}{(-1)(l \sin \alpha + x) \sin \alpha} - \frac{x}{(l \sin \alpha + x) \sin \alpha} \end{aligned}$$

After final simplifications, the slope becomes

$$m_{DP} = -\cot \alpha - \frac{x}{(l \sin \alpha + x) \sin \alpha}$$

We also know that the slope of line segment AD is

$$m_{AD} = \frac{-l \sin \alpha - 0}{l \cos \alpha - 0} = -\cot \alpha$$

Since α is an acute angle, $\sin \alpha > 0$. Since x and l are also positive, we see from the above expressions that the slope of DP is indeed more negative than the slope of AD . Q.E.D.

THE BUSY BEAVER FUNCTION

Dmitriy Aronov

Many functions that we often deal with approach infinity. The definition of such behavior is relatively simple – we say that function f approaches infinity if for any M , no matter how large, there exists a “cutoff” value N , such that $f(n) > M$ for all $n > N$. For instance, given the function $f(x) = x^2$ and $M = 9$, we can easily find the cutoff value $N = 3$, because $n^2 > 9$ for all $n > 3$. In general, this function approaches infinity because, given any value M , the cutoff clearly exists and is equal to $\sqrt{M}$. When dealing with such functions, it is also important to know how fast they approach infinity. We say that f approaches infinity faster than g if

$$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \infty$$

For example, x approaches infinity faster than $\sqrt{x}$, which in turn approaches infinity faster than $\ln(x)$. Polynomial functions of higher degrees grow faster than those of lower degrees, while exponential functions approach infinity faster than polynomials. Looking for other record holders, we find the factorial, $x!$, and an even faster growing partition function, $p(x)$, defined as the number of ways a positive integer can be written as a sum of positive integers. We thus have a list of several classes of functions, where each class dominates all previous ones:

$$\log_a x < x^a < a^x < x! < p(x)$$

Here, a is a positive number and the symbol $<$ is used to compare classes. The list is certainly not complete and includes only the most common classes of functions. We can also combine classes to create an infinite variety of functions. For example, $2^{x!}$ is a combination of $x!$ and a^x with $a = 2$.

Having this in mind, we can get to the main topic of this article:

Is there a function which grows faster than any other function?

Suppose such function exists, and let it be $f(x)$. There is a contradiction right away because $[f(x)]^x$ grows even faster. However, we can disregard this argument by noting that $[f(x)]^x$ is a combination of functions from different classes. What we are really interested in is the class of functions which would go last in the above list. Such function indeed exists and was found by Tibor Rado in 1962. To define it, we first need to introduce Turing machines.

A Turing machine consists of three components: 1) a processor, which at any given time is in exactly one of the n possible states, 2) memory in the form of an infinitely long tape containing a sequence of 0's and 1's, 3) a pointer, which at any given time points at some position on the tape. In addition to these “hardware” components, there is a program, which tells the machine what to do. It is a set of statements in the form $\{A, B, C, D, E\}$, which means:

If the current state of the processor is A and the current symbol on the memory tape is B, then write C on the tape, move the pointer 1 unit in the direction D, and set the processor to state E. Here, A is the number of the state (from 1 to n), B and C are binary digits, D is either “left” or “right,” and E can be either a state from 1 to n or “halt.” The command “halt” tells the machine

to stop operating. One can easily create programs in which the machine never halts. Obviously, only those machines that eventually halt can have a practical meaning.

Although they are based on simple principles, Turing machines can behave in very complex ways. This holds true even for machines with small numbers of states (fig. 1). Some Turing machines halt immediately; others go off to infinity on the memory tape. Some have periodic behaviors; many others “jump around” chaotically before their fate becomes evident.

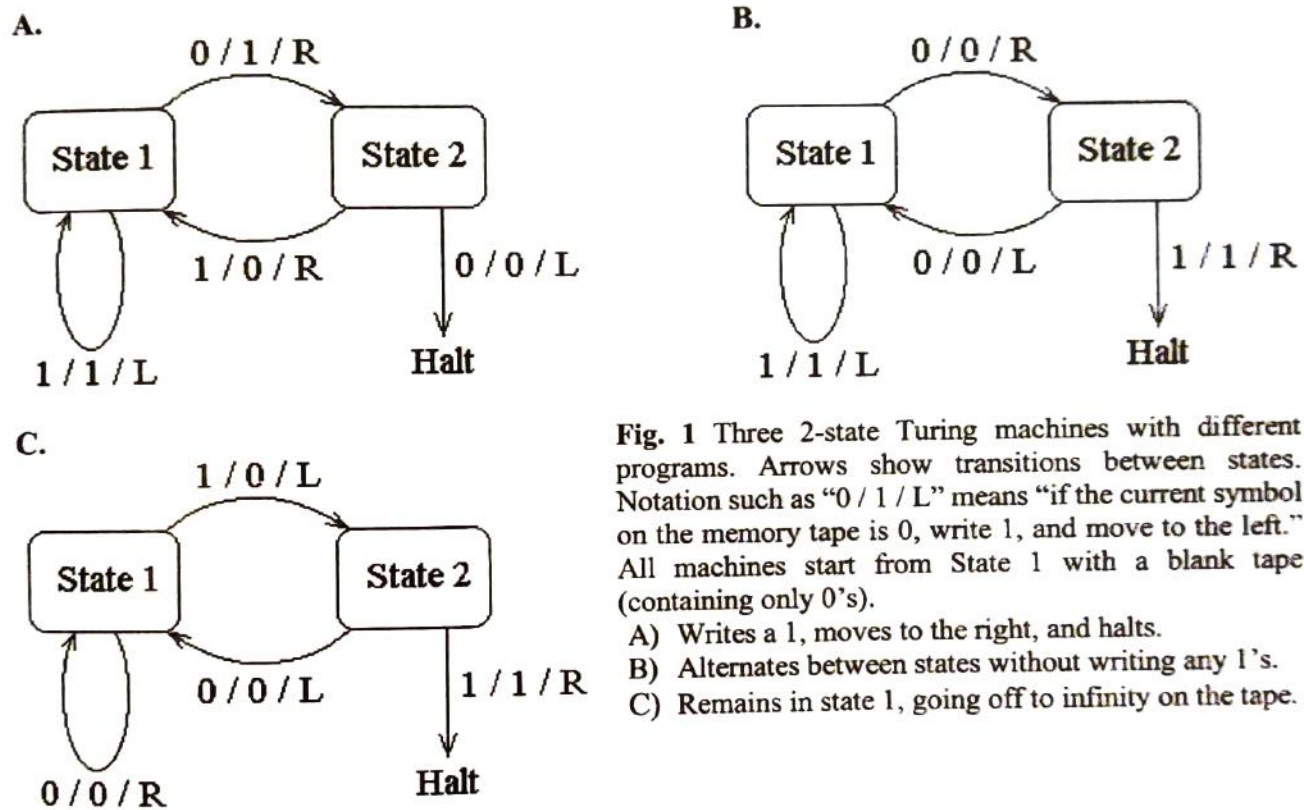


Fig. 1 Three 2-state Turing machines with different programs. Arrows show transitions between states. Notation such as “0 / 1 / L” means “if the current symbol on the memory tape is 0, write 1, and move to the left.” All machines start from State 1 with a blank tape (containing only 0’s).

- A) Writes a 1, moves to the right, and halts.
- B) Alternates between states without writing any 1’s.
- C) Remains in state 1, going off to infinity on the tape.

A Turing machine is the most generalized representation of a device that can perform mathematical and logical operations. In fact, given a sufficient number of states, one can create a Turing machine whose performance is identical to the performance of any other computer, no matter how complex. A 2-state machine, for instance, can perform addition, while a 16-state can perform multiplication. Given more states, one can program the machine to do more complex operations, such as determining whether a number is prime or composite with 39 states [2]. Turing machines operate in the *unary* system, where an integer n is represented by a sequence of n ones. This is necessary in order to make Turing machines as simple as possible, so that any other computer can be expressed in terms of them. To perform calculations, the machine starts on the input tape and halts after writing the output.

Tibor Rado imagined a beaver running busily back and forth on the tape, putting 1’s in various places. He asked how many 1’s the busiest beaver would put. The answer, which he accordingly called the *busy beaver function*, has been the subject of extensive research in the last four decades. First, we need to state the formal definition.

Definition. *Busy beaver function*, $\Sigma(n)$, is the maximum number of 1’s that an n -state Turing machine can output if it starts from a blank tape (containing only 0’s) and eventually halts.

It is clear why such function would have practical and theoretical importance. Since Turing machines output numbers in unary format, the busy beaver function values set a limit to the size of the output that a machine can provide. In computation theory, this is closely related to the complexity of calculations that a machine can perform.

Much of the research related to Turing machines attempted to calculate the busy beaver function values for small values of n . The only method of doing this is to actually look through all possible Turing machines with a given number of states. It is not hard to work out that the total number of n -state Turing machines is $(4n)^{2n}$, so the amount of time it takes to do these calculations increases very rapidly. Thus, it is crucial to eliminate “lazy beavers” and non-halting machines as early as possible in the calculation to save time. This is a challenging problem because Turing machines can behave in very complex ways before halting, becoming periodic, or going off to infinity.

After defining the function, Rado published its first three values (table 1). He found that $\Sigma(1)=1$, $\Sigma(2)=4$, and $\Sigma(3)=6$. Later, with faster computers and improved algorithms, Brady calculated the next value to be $\Sigma(4)=13$. Consequent values gave mathematicians a lot of trouble because of the increasingly complex behavior of many machines and the extremely fast growth of the function values. In fact, only recently Heiner Marxen found 4098 to be the function’s 5th value. The value of $\Sigma(6)$ makes an even higher jump. By now, mathematicians have only looked through about 90% of all 6-state Turing machines, so the final answer is not certain. The champion machine known so far prints more than 6.4×10^{462} 1’s on the memory tape (fig. 2). This is far beyond the number of elementary particles in our universe!

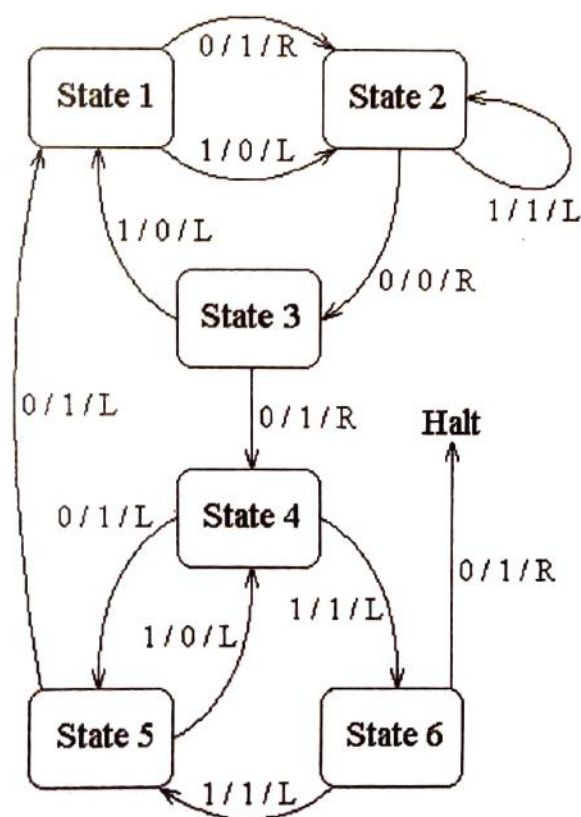


Fig. 2 Current Turing machine champion for the 6-state busy beaver function [1]. Starting from State 1, it halts after writing more than 6.4×10^{462} 1’s on the memory tape.

n	$\Sigma(n)$
1	1
2	4
3	6
4	13
5	4098
6	$> 6.4 \times 10^{462}$
...	...
12	$> 6 \cdot 4096^{4096^{4096 \dots 166 \text{ times}}}$

Table 1. Known busy beaver function values as well as a couple of lower bounds.

After seeing such a breathtaking growth of values, it may not be surprising to learn that the busy beaver function indeed grows faster than any computable function. By computable functions, we mean those that can be calculated with Turing machines, which include pretty much all function humans ever have to deal with. Here, we present a proof of this phenomenon.

Theorem. The busy beaver function, $\Sigma(x)$ approaches infinity faster than any computable function. That is, $\lim[\Sigma(x)/f(x)] = \infty$ for all computable functions f .

Proof. Suppose we have an arbitrary computable function f . We define another function:

$$F(x) = \sum_{i=0}^x [f(i) + i^2]$$

Since f is computable, so is F . Thus, we can create a Turing machine M_F that starts with a block of x 1's on the tape and outputs a block of $F(x)$ 1's, so that the blocks are separated by at least one zero (this is essentially how Turing machines calculate functions). Suppose that M_F has k states.

Now, consider a Turing machine M , which starts with a blank tape (all zeros), and writes a block of x 1's. This operation can be easily done with x states. Next, this machine takes the block it just wrote as the input and mimics M_F by writing a block of $F(x)$ 1's. As we already said, this operation can be done with k states. Finally, our machine takes the second block it wrote and mimics M_F again, this time calculating $F(F(x))$. This step again requires k states. Thus, our machine has $x + 2k$ states and writes $x + F(x) + F(F(x))$ 1's on the initially blank tape. Since $\Sigma(x + 2k)$, by definition, is the maximum possible number of 1's that could be written by this machine, we can say that

$$\Sigma(x + 2k) \geq x + F(x) + F(F(x))$$

From its definition, $F(x) \geq x^2$. We know that x^2 eventually becomes greater than any linear function, so there exists some constant c , such that $x^2 > x + 2k$ for all $x \geq c$. It follows that $F(x) > x + 2k$ for all $x \geq c$. Now, from the definition of F , we know that it is an increasing function. Thus, $F(F(x)) > F(x + 2k)$ for all $x \geq c$. We also know that $F(x) > f(x)$ for all positive x . It follows that

$$\Sigma(x + 2k) \geq x + F(x) + F(F(x)) > F(F(x)) > F(x + 2k) > f(x + 2k)$$

for all $x \geq c$. Thus, $\Sigma(x)$ will eventually be greater than any arbitrary function. QED.

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MAGIC SQUARES

Minggang Guo

What is a magic square? In a pure or normal magic square, all rows, columns, and the two main diagonals must sum to the same value and the numbers used must be consecutive from 1 to n^2 , where n is the number of cells on each side of the square. Magic squares have provided many with entertainment and challenge. The magic square seems to be of Chinese origin, since it was mentioned in a manuscript called *lo shu* from the time of Emperor Yu around 2200 B.C. In the western world, the work of Theon of Symrna in 130 A.D. first mentioned the magic square. In the 9th century, Arab astrologers used magic squares in horoscope calculations. Finally, during the Renaissance period, magic squares through the works of Greek mathematician Moschopoulos spread to the Western Hemisphere. They were also frequently found in various cultures, such as Egypt and India, engraved on stone or metal and worn as talismans. Thos originated from a belief that magic squares have astrological and divinatory qualities. Later, many mathematicians and artists such as Kraitchik, A. W. Johnson, Jr., and Albrecht Dürer have found ways to construct various types of magic squares.

The simplest way of making odd order magic squares, or magic squares with odd number of cells on each side, is the staircase method: 1) Start with the number 1 in the top-center box of the square. 2) Place the next number in the space located to the top-right of the current space. If the box is not in the square, attach an imaginary square in the necessary place and find the location of the box there. 3) If the box at the top-right is occupied, place the number in the box immediately below the previous number. 4) Repeat steps 2 and 3 until you fill the entire magic square. Here are the steps of making a magic square with order 3:

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The rule is harder to generalize for even order magic squares. Instead, there are methods like the diagonal method, which applies only to a 4×4 magic square. Sequentially place the numbers in the rows of the magic square. Draw two diagonals of the square. If a number lands on one of the diagonals, its location must be switched with its complement that adds to 17.

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There is an interesting approach to creating magic squares which takes an existing magic square and surrounds it with a magic border. Any one of the 5th order magic borders below is easily constructed by a trial-and-error method.

20	1	21	19	4
2				24
3				23
18				8
22	25	5	7	6

2	25	23	7	8
20				6
21				5
4				22
18	1	3	19	24

4	25	7	23	6
2				24
21				5
18				8
20	1	19	3	22

Notice that the sum of opposite entries and the sum of opposite corners is 26, or $n^2 + 1$ in general. Also, numbers 9 through 17 are not used for the border. To make a border, the top row and the right hand column have to be selected to give the magic constant 65 (see formula below). Then complementary pairs on the bottom and left-hand side can be filled out. The equation for the magic sum of a magic square is

$$S = \frac{1}{n} \sum_{k=1}^{n^2} k = \frac{1}{2} n \cdot (n^2 + 1)$$

where n is the order of the magic square, or the number of cells on each side. Thus, the magic sum is 15 for a 3×3 square and 65 for a 5×5 square. After the magic border is added to a 3×3 square, the sum becomes $15 + 26 = 41$. Thus, we need to increase the sum by 24 in order to have a magic sum of 65. This can be done by adding 8 to each entry of the original 3×3 square. The numbers in this square will thus be 9 through 17 instead of 1 through 9, and we reserved these numbers earlier. The resulting square looks like this:

20	1	21	19	4
2	16	9	14	24
3	11	13	15	23
18	12	17	10	8
22	25	5	7	6

There are a lot of different ways to construct magic squares, such as algebraic methods, various equations, and methods like the magic border. In addition, there are geometric magic squares where the products of the sides are equal, magic cubes, magic circles, spheres, and stars. These form a very large and interesting field of study.

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FERMAT NUMBERS AND GAUSSIAN POLYGONS

Xunlei Li

Number theory is a branch of mathematics that deals with the properties and relationships of numbers. It has been one of the most studied topics in the history of mathematics. From the time of Euclid to the modern age, people have tried to understand the relationships between numbers, especially integers. Among integers, prime numbers have been the subject of much research. In the past few centuries, many famous mathematicians carefully studied whether numbers in special forms are primes. One of the most famous mathematicians who devoted time to this was Pierre de Fermat. Although numbers in the special forms that he discovered were not all primes, they did find their own unique use in the field of straight edge and compass geometric constructions.

Numbers in the form of $2^m + 1$, where m is a natural number, were studied long ago. Although they are not all prime numbers, they are all odd. These numbers caught the attention of Fermat. Fermat conjectured that in order for these special numbers to be prime, m must be of the form 2^n . These numbers were hereafter named Fermat numbers. F_n is used to denote them.

Thus, the general form of a Fermat number would be $F_n = 2^{2^n}$. Fermat thought that all numbers in this form were prime but could not prove it. The first five numbers of this form were found to be prime by Fermat himself. These numbers, $F_0 = 3$, $F_1 = 5$, $F_2 = 17$, $F_3 = 257$, and $F_4 = 65537$, are all primes. Since F_5 has 10 digits, in order to test its primality, it would be necessary to have a prime factor table that goes up to 100,000 or some criterion for a number to be the factor of F_n . Neither option was available to Fermat, and thus he was unable to test other Fermat numbers for primality.

An interesting property of Fermat numbers is that they are all relatively prime to one another. Let's suppose that F_n and F_{n+k} , where $k > 0$, are two Fermat numbers and that both F_n and F_{n+k} are divisible by M . If $x = 2^{2^n}$ then

$$\frac{F_{n+k} - 2}{F_n} = \frac{2^{2^{n+k}} + 1 - 2}{2^{2^n} + 1} = \frac{x^{2^k} - 1}{x + 1} = x^{2^k-1} - x^{2^k-2} + x^{2^k-3} - \dots - 1$$

This shows that $F_{n+k} - 2$ is divisible by F_n . Thus, M is a factor of both of them. Since M is a factor of both F_{n+k} and $F_{n+k} - 2$, M must be a factor of 2. Since all Fermat numbers are odd, M must be 1. Thus, the common divisor of two different Fermat numbers is never greater than 1, making them relatively prime.

Later mathematicians found new ways of determining the primality of Fermat numbers. Euler discovered that every factor of F_n , where $n \geq 2$, must be of the form $k \cdot 2^{n+2} + 1$. Using this discovery, he was able to prove that F_5 is not prime. Instead, it has a factor of 641.

Fermat numbers had for the most part been counted as part of pure math. However, a German mathematician Carl Gauss was able to use them in regular polygonal constructions with a compass and straightedge. Gauss was able to prove that the only constructible regular polygons with a prime number of sides are those that have a Fermat prime number of sides. For example, a

triangle has 3 sides, a pentagon has 5 sides, and a heptadecagon has 17 sides. They are all constructible because 3, 5, and 17 are Fermat primes.

Despite all the advances that the modern supercomputers have brought us in the understanding of Fermat numbers, we still know very little about them. Some of the “small” Fermat numbers have yet to be determined to be composite or prime. F_{22} , F_{24} , and F_{28} are the smallest Fermat numbers that are still unsolved. Despite the fact that Fermat’s conjecture is false, there are still many questions left.

No one has yet proven that there are only a limited number of primes among Fermat numbers. A German mathematician Eisentein proposed in 1844 that there are in fact infinitely many Fermat primes. An anonymous writer stated in 1828 that the numbers $2+1$, 2^2+1 , $2^{2^2}+1$, ... are all primes. However, this was proven false after Selfridges, found F_{16} to be composite. It is very probable that the last Fermat prime is indeed $F_4 = 65537$ and that this is the limit of the constructions that Gauss discovered.

Another interesting conjecture that came out of the Fermat numbers is that every Fermat number is square free (that is, without any square factors other than 1). This was first proposed by Lehmer and Schnizle. However, this fact is not proved (or disproved) yet, and Fermat numbers continue to pose many questions to number theory.

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RUBIK'S CUBE

Frankie Chen

The Rubik's Cube has been called "the perfect puzzle" and "the best puzzle ever." With just a few turns, the small colored cubes of the Rubik's Cube become thoroughly mixed up. The purpose of the puzzle is to return every side of the cube to a solid color (solid state of the cube). What distinguishes the Rubik's Cube from other puzzles is that its physical structure lets mathematicians talk about abstract concepts, which must otherwise be presented with only theoretical examples.

Group theory includes the study of permutations and is therefore basic in coding theory, cryptography, card-shuffling tricks used by magicians, etc. As another example, the rigid motions of space form a group, making group theory crucial in theoretical physics. The concepts of the Rubik's Cube are important in clarifying other fields of mathematics. For example, using the Rubik's Cube as a model of group theory, we can study the solvability of polynomial equations and the structure of geometric and topological objects.

A group is made up of two things: a set of objects, denoted S , and an operation, denoted $*$, which combines two of these objects together to form another object in S . For example, the set of objects S could be real numbers and the operation for combining them could be addition or multiplication. In the case of the cube, the set of objects is all possible permutations on the cube. Note, that each permutation can be defined by the rotations that are necessary to get to it from the cube's solid state. The group operation for combining permutations is "followed by". Therefore, if X and Y are permutations on the cube, then X and Y are combined to form a single permutation X followed by Y , which can be written simply as XY . Essentially, XY is the permutation formed when one starts from the solid state of the cube, does all rotations to get to state X , and then performs all rotations that would be necessary to get to Y from the solid state. Clearly, the inverse of X , X^{-1} is formed when one starts from the solid state and does all rotations that would be necessary to get to the solid state from X . The identity element of the group, I , is clearly the solid state of the cube.

The case in which the group S is finite is especially interesting with regard to the Rubik's Cube. Finite groups, of which the permutations on the cube are an example, have some unique properties. For example, if X is the permutation of a process on the cube, then for some positive integer n , we would have the permutation $X^n = I$, where I is the identity permutation. The value of n which satisfies this is called the order of X . Similarly, the set $\langle X \rangle = \{X, X^2, X^3, \dots, X^n, \dots\}$ can only have a finite number of elements since it is a subset of the finite group S . Choose n to be the smallest number such that $X^{n+1} = X^k$ with $0 < k \leq n$. In other words, X^{n+1} is the first element in the sequence which equals an earlier element.

It is easy to prove that k must equal 1. If $k > 1$, we would have $X^n = X^{n+1} * X^{-1} = X^k * X^{-1} = X^{k-1}$, and for all $k > 1$, X^{k-1} would be in the set $\langle X \rangle$. Thus, X^n would be an element in our sequence which equals an earlier one. This is a contradiction to our initial assumption that X^{n+1} was the first such element. Therefore, the assumption that $k > 1$ is false, and k must be 1. It is important to realize that the identity and the inverse of each element is in $\langle X \rangle$, since $X^n = X^{n+1} * X^{-1} = X * X^{-1} = I$ and $X^{n-1} = X^n * X^{-1} = I * X^{-1} = X^{-1}$.

From this, it is concluded that $\langle X \rangle$ is the group generated by X . Any finite group that can be generated by a single element is called a cyclic group. We say that $\langle X \rangle$ is a cyclic group of order n . It is important to note that $X^m = I$ if and only if m is divisible by n . If X is an element of a finite group S , then $\langle X \rangle$ is a cyclic group, and the element X^{-1} generates the same cyclic group. However, it is not necessarily true that for every k , X^k will generate the same cyclic group that X does.

One interesting question to consider is this: how many different permutations of the cube are there, including different flips and twists of pieces as different permutations but not distinguishing between different orientations of the cube as a whole? To calculate the order of the cube group, first consider how many permutations of the eight corner locations are possible if the twists are ignored. The total number of possible permutations of a set of n objects in n locations is $n!$, or n factorial. Since every permutation can be decomposed into successive pair exchanges, every permutation of corner locations is possible on the cube. Therefore, we have $8! = 40,320$ possible permutations of corners on the cube. Similarly, there are $12! = 479,001,600$ possible permutations of edge locations on the cube.

However, not all permutations of corner locations can go with all permutations of edge locations. The permutation of corners and edges together must be even, so that even corner permutations must occur with even edge permutations and odd with odd. Every even corner permutation can occur with each even edge permutation. In addition, every odd corner permutation can occur with each odd edge permutation. Since half the permutations are odd and half are even, the total number of permutations of the cube, not counting twists or flips, is $(8! \cdot 12!)/2 = 9,656,672,256,000$.

Now, twists and flips have to be considered. Any corner can be twisted in three orientations, except for the last one, whose orientation is fixed by the other seven. Similarly, any of the edges can be flipped in two ways, except for the last, whose orientation is determined by the first eleven. Therefore, each permutation of the cubes can have $3^8/3 = 2,187$ corner orientations and $2^{12}/2 = 2,048$ edge orientations.

Therefore, the total number of cube permutations counting twists and flips is

$$\frac{8! \cdot 12!}{2} \cdot \frac{3^8}{3} \cdot \frac{2^{12}}{2} = 43,252,003,274,489,856,000$$

In scientific notation, this number is $4.3 \times 10^{19} = 43$ quintillion, and is the order of the entire Cube Group. With so many different combinations, it is no wonder that the Rubik's Cube is such a thrilling mathematical puzzle!

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BUFFON'S NEEDLE EXPERIMENT

William Chen

Buffon's needle experiment is one of the oldest problems in geometric probability. It involves dropping a needle on a lined sheet of paper with equally spaced parallel lines. The goal is to determine the probability that the needle intersects a line. Before the needle experiment, Buffon considered the probability of dropping a coin of radius r onto a square of side s and having the coin land completely within the square (refer to fig. 1). Buffon reasoned that the coin would fall completely within the square if and only if the center of the circle landed within an imaginary square that was drawn within the original square. The sides of this imaginary square are $s - 2r$ in length. The ratio of the area of the smaller square, $(s - 2r)^2$, to the area of the larger square, s^2 , is the solution to the problem. This simple problem nicely illustrates the usefulness of geometry in certain probability problems.

Fig. 1

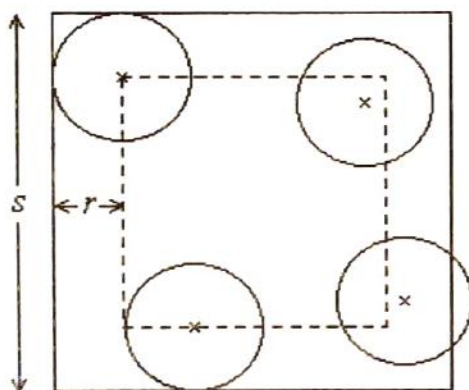
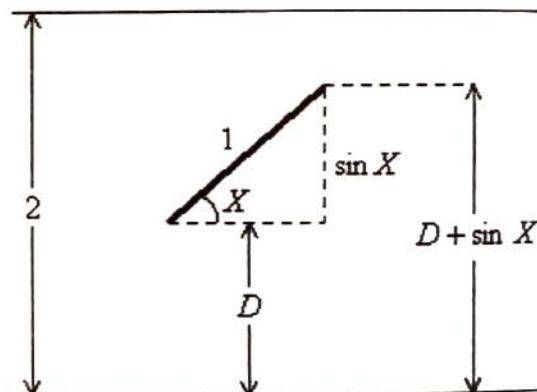


Fig. 2

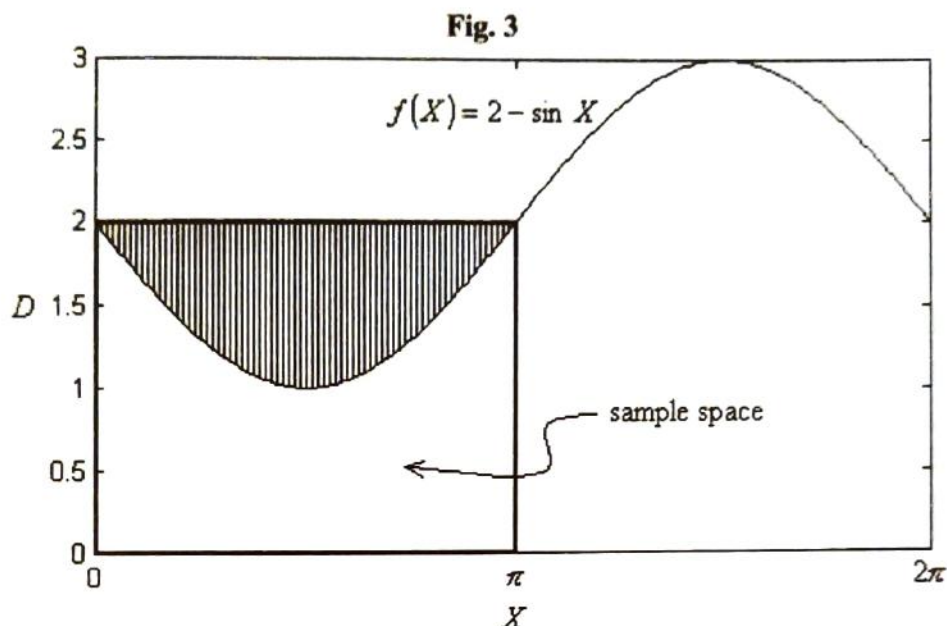


Let's now solve a specific case of Buffon's needle experiment, where the lines on the paper are 2 units apart and the needle is 1 unit long. Using the lowest end of the needle as our reference point, let D be the distance from that point to the line beneath it (not necessarily the closer line) and let X be the angle from the horizontal at which the needle falls (refer to fig. 2). The higher end of the needle is $D + \sin X$ units from the line beneath the needle. From fig. 2, we see that as long as $D + \sin X$ is greater than or equal to 2, the needle will intersect a line. We know that D is in the interval $[0, 2)$ and X is in the interval $[0, \pi)$. The entire sample space of the experiment can be represented by a rectangle on a coordinate plane with base π and height 2 (fig. 3). On the same coordinate plane, we also plot the function $f(X) = 2 - \sin X$, with X along the horizontal axis and D along the vertical axis.

We have already said that in order for the needle to intersect a line, the condition $D + \sin X \geq 2$ must be satisfied. This is equivalent to $D \geq 2 - \sin X$, which corresponds to the shaded region above the graph of $f(X)$ on our coordinate plane. The area of this region is represented by the expression

$$A_{\text{shaded}} = \int_0^{\pi} [2 - (2 - \sin X)] dX = \int_0^{\pi} \sin X dX = 1$$

The area of the entire sample space rectangle is $1 \cdot \pi = \pi$. By the same principle as before, we know that the probability of intersecting a line is the ratio of the two areas, or $1/\pi$.



This conclusion is interesting because it provides a crude statistical way to approximate π . To do the approximation, simply take the number of needles dropped and divide by the number of needles that intersected a line. This application of the problem can be easily verified with a computer program that simulates the dropping of a needle numerous times. The following chart shows the results of such simulation:

Number of Drops	Number of Intersections	Approximate Value of π	Error (%)
1	0	∞	∞
10	5	2	36.338
100	33	3.0303	3.6426
1000	311	3.2154	2.3494
10000	3150	3.1746	1.0507
100000	32023	3.1228	0.5982
1000000	318394	3.1408	0.0252

One can expand on this problem to many other geometric shapes. A particularly interesting problem is what happens if the length of the needle is twice the length of the distance between the lines. What is the probability that the needle will intersect 1 line? 2 lines?

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WEBSITE REVIEW: DR. MATH

Annabelle Lee

If you've ever had a problem with homework and didn't know where to turn, Dr. Math could have helped you. Dr. Math is an online question and answer service, staffed by volunteers from around the world. It got its start in 1994 when Swarthmore college students who were interested in helping other students brought it to life. There was only a handful at first, but now there are over three hundred, "math doctors," from around the globe, including several students at Stuyvesant High School.

The querying process begins when a student accesses the Math Forum's site at <http://forum.swarthmore.edu> and goes to Ask Dr. Math. A screen loads with many options: looking at answers to the most frequently asked questions, a search form to look through archived responses to a multitude of questions ranging from beginning fractions to differential equations, and information divided by school level. There is also a special area devoted to helping math teachers instruct students.

It is only after a student has utilized the informational resources available on the site that he should submit a question. The query in fact is specifically designed to make sure that a student has indeed used all other options. You might think this is silly, but it is necessary. Ask Dr. Math receives over three hundred questions a day, and it is difficult for the Math Doctors, who have so many other commitments, to answer all the questions. In fact, Dr. Math currently answers about half of the incoming questions.

When you finally submit your query, which also asks your email address and age so that the answer is delivered right to your email address and tailored to your age group, it is sent to the Dr. Math Triage area. When it is in the Triage area, any tenured math doctor can answer it. It is on a long list that includes all the question sent during the previous seventy two hours, and questions are not answered in any systematic way but only as the doctors choose according to their inclinations on any given day.

Overall, using the Ask Dr. Math Service is a positive experience, even if your question isn't answered (and they really do their best to answer as many as possible). If you're interested in mathematics, any branch at all, you're bound to learn something new by reading the archives, even if it's not the particular item you wanted. Furthermore, if you do receive an answer it will be easy to understand and written specifically for you and in answer to your question. There's only one thing you may not like about this site: they don't give out homework answers. It's still a really great resource, and you should definitely take advantage of it if you ever need help with your math homework or even if you're just looking for a new area of mathematics to explore.

The Origin of "Elusive Formulas"

When two very bored math nerd wannabes get drunk, the result is what you see in the next ten pages. The project all started when Ming Jack Po (Stuyvesant 2001 hopeful) panicked before the AMC (which to add to the humiliation, he did very badly on throughout his high school career) and accidentally stumbled upon IMSA's "Noah Sheets". Upon further inquiries, Jack finds out that not only does IMSA have a cheat sheet, so does Bronx Science! In fact, Bronx Science was so brazen as to publish their cheat sheet in their math magazine, "Math Bulletin", one copy which currently resides in Jack's basement gathering dust. So, Jack being the amazingly bright guy that he is, came up with the brilliantly original idea of creating Stuyvesant's very own cheat sheets. And here is where Kevin Zheng, also a Stuyvesant 2001 hopeful, enters the picture. Kevin, another self proclaimed math as well as computer nerd quickly declared that this task will be nothing more than child's play. What followed is two months of painful yet expected procrastination that all culminated to a weekend of grueling work before the deadline. During the weekend, Kevin pumped out diagrams after diagrams, while Jack whined about one diagram after the other (um, not to say that Jack didn't do any actual work, um... yes). After hours of painstaking labor, the "Elusive Formulas" slowly morphed into the pages you see before you.

Both Kevin and Jack have competed in many math competitions throughout their high school careers, and know of most of these formulas collectively as "formulas that I should have remembered during the _____ competition", hence the name "Elusive Formulas". It is our hopes that you will find these sheets useful in your math classes, or math team training, or prying open a window in your basement.

Finally, we would like to thank Jan Siwanowicz, Jeff Amlin, Kamaldeep Gandhi and especially Joel Lewis for informing us that half these formulas were either wrong or made up by Jack (which btw is a necessary and sufficient condition for proving that they are wrong). Thanks guys!

- The radius of the inscribed circle is r .
- The radius of the circumscribed circle is R . (not shown here)
- The area of the polygon is K .
- The semi-perimeter of the polygon is S .
- The altitude to sides a, b, c are h_a, h_b, h_c respectively.
- The angle bisectors to side a, b, c are t_a, t_b, t_c respectively.

If you still find any errors in, or have any suggestions for "Elusive Formulas" please contact Mr. Daniel Jaye, Assistant Principal of the Department of Mathematics at Stuyvesant High School, 345 Chambers Street, New York, NY 10282-1099. Phone: (212) 312-4859 Fax: (212) 312-4858 Email: jayed@stuy.edu

Section A – Algebra

If $\frac{a}{b} = \frac{c}{d}$ then:

$$\frac{a+b}{b} = \frac{c+d}{d}$$

$$\frac{a+b}{a-b} = \frac{c+d}{c-d}$$

$$\frac{a}{b} = \frac{c}{d} = \frac{a+c}{b+d}$$

- $(a+b)^3 = a^3 + b^3$ iff $a = 0$ or $b = 0$ or $(a+b) = 0$
- $a^3 \pm b^3 = (a \pm b)(a^2 \mp ab + b^2)$
- $x^n \pm y^n = (x \pm y)(x^{n-1} \mp x^{n-2}y + x^{n-3}y^2 + \dots \mp xy^{n-2} + y^{n-1})$ [$x^n + y^n$ is only true for odd n .]
- $(x \pm y)^n = \binom{n}{0}x^n \pm \binom{n}{1}x^{n-1}y + \binom{n}{2}x^{n-2}y^2 \pm \dots$

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_0 x^0 = 0$, then

Sum of roots taken one at a time
(the sum of the roots)

$$\sum_{i=1}^n r_i = -\frac{a_{n-1}}{a_n}$$

Sum of roots taken two at a time

$$\sum_{i=1}^n r_i r_j = \frac{a_{n-2}}{a_n}$$

Sum of roots taken p at a time

$$\sum_{i=1}^n r_i r_j \dots r_k = (-1)^p \frac{a_{n-p}}{a_n}$$

Arithmetic Progression :

If $a_1, a_2, a_3, \dots, a_n$ are in arithmetic progression with common difference d :

n^{th} term of an arithmetic progression

$$a_n = a_1 + (n-1)d$$

Sum of an arithmetic progression up to term n

$$\sum_{i=1}^n a_i = \frac{n}{2}[a_1 + a_n] = \frac{n}{2}[2a_1 + (n-1)d]$$

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}$$

Geometric Progression :

If $a_1, a_2, a_3, \dots, a_n$ are in geometric progression with common ratio r :

n^{th} term of a geometric progression

$$a_n = a_1 r^{n-1}$$

Sum of a geometric progression up to term n

$$\sum_{i=1}^n a_i = \frac{a_1(1-r^n)}{1-r}$$

Sum of an infinite geometric progression

$$\sum_{i=1}^{\infty} a_i = \frac{a_1}{1-r} \text{ iff } |r| < 1$$

- For $P(x)$, $(x-r)$ divides $P(x)$ iff $P(r)=0$ where r is a real or complex number
- $P(x)$ divides the polynomial $Q(x)$ iff every root of $P(x) = 0$ is also a root of $Q(x) = 0$ (if a root of $P(x)$ has multiplicity of N , then the root must also have at least a multiplicity of N in $Q(x)$)

Rational Root theorem

If $P(x) = a_n x^n + a_{n-1} x^{n-1} + a_{n-2} x^{n-2} + \dots + a_1 x + a_0$ is a polynomial with integer coefficients and $\frac{p}{q}$ is a rational root of the equation $P(x) = 0$ (where p and q are relatively prime), then p must divide a_0 , while q must divide a_n .

- If $P(x)$ is a polynomial with real coefficients and $P(a + bi) = 0$, then $P(a - bi) = 0$.
- If $P(x)$ is a polynomial with rational coefficients and $P(a + b\sqrt{c}) = 0$, then $P(a - b\sqrt{c}) = 0$.

Section B – Number Theory

Pythagorean Triples:

All positive integer solutions to the equation $a^2 + b^2 = c^2$ are given by:

$$a = k(m^2 - n^2)$$

$$b = k(2 * m * n)$$

$$c = k(m^2 + n^2)$$

Where m , n , and k are arbitrary positive integers, with $m > n$. For relatively prime triples (a, b, c) , we need $k=1$, and m and n relatively prime and of opposite parity.

Fermat's Last Theorem:

The equation $a^n + b^n = c^n$ has no positive integral solutions for $n > 2$.

Fermat's Little Theorem:

If p is prime and a is relatively prime to p , then $a^{p-1} - 1$ is a multiple of p .

Section C – Logarithms

$\log_b N = P \Leftrightarrow b^P = N$	$\log_b N = \frac{\log_a N}{\log_a b}$	$\log \frac{m * n}{q} = \log m + \log n - \log q$
$\log N^p = p \log N$	$\log_b a = \frac{1}{\log_a b}$	$\log_a b * \log_b c = \log_a c$

Section D – Point-Line Distance

Between (x_0, y_0) and line $ax+by+c=0$	$\frac{ a*x_0 + b*y_0 + c }{\sqrt{a^2 + b^2}}$
Between (x_0, y_0, z_0) and line $ax+by+cz+d=0$	$\frac{ a*x_0 + b*y_0 + c*z_0 + d }{\sqrt{a^2 + b^2 + c^2}}$

Section E – Inequalities

- If $x > 0$, $f(x) = x + \frac{1}{x}$ attains its minimum value when $x = 1$
- If $a + b$ remains constant, the product $a*b$ is largest when $a=b$.
- The “arithmetic-geometric-harmonic mean inequality:” for $a, b > 0$, $\frac{a+b}{2} \geq \sqrt{ab} \geq \frac{2}{\frac{1}{a} + \frac{1}{b}}$,

with equality holding iff $a=b$. In general for $a_1, a_2, \dots, a_n \geq 0$,

$$\frac{1}{n}(a_1 + a_2 + \dots + a_n) \geq \sqrt[n]{a_1 * a_2 * \dots * a_n} \geq \frac{n}{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}, \text{ with equality holding iff}$$

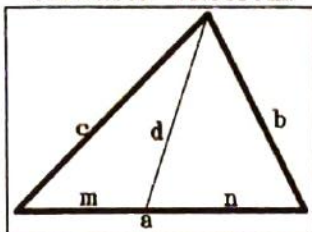
$$a_1 = a_2 = \dots = a_n.$$

Section F – Divisibility Rules

Divisor	Criterion
3	The sum of the digits of N is divisible by 3
4	The last two digits of N is divisible by 4
7	Take the last digit of N, double it, and then subtract it from the rest of the number, if you get 0 or an answer divisible by 7, then N is divisible by 7.
8	Last 3 digits of N is divisible by 8.
9	The sum of the digits of N is divisible by 9.
11	The difference between the sum of the first, third, fifth ... digits of N and second, fourth, sixth.. digits of N is either 0 or divisible by 11, then so is N.
13	Delete the last digit of N. Then subtract nine time the deleted digit from the remaining number. If what is left is divisible by 13, then so is N.

Section G – The Triangle

Stewarts' Theorem

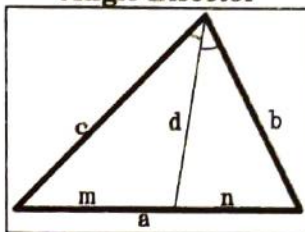


$$man + dad = bmb + cnc$$

$$(man + dad =$$

$$\text{bomb} + \text{sink})$$

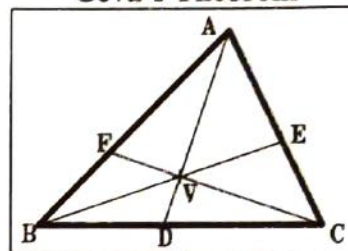
Angle Bisector



$$\frac{c}{b} = \frac{m}{n}$$

$$d^2 = bc - mn$$

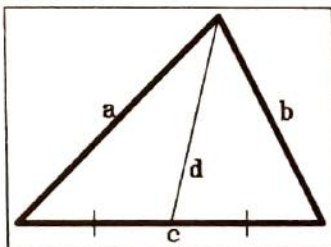
Ceva's Theorem



$$\frac{AF}{FB} \cdot \frac{BD}{DC} \cdot \frac{CE}{EA} = 1$$

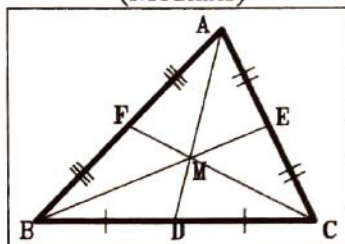
$$\frac{VD}{AD} + \frac{VE}{BE} + \frac{VF}{CF} = 1$$

Medians



$$d^2 = \frac{1}{2}(a^2 + b^2) - \frac{1}{4}c^2$$

Centroid (Medians)

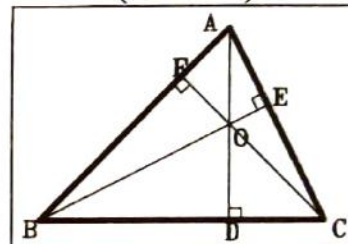


$$\frac{AM}{MD} = \frac{BM}{ME} = \frac{CM}{MF} = 2$$

$$K_{AFM} = K_{FBM} = K_{BDM} = K_{DCM}$$

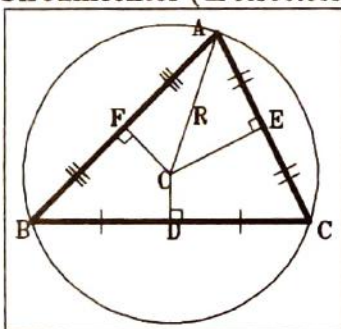
$$= K_{CEM} = K_{EAM} = \frac{1}{6}K_{ABC}$$

Orthocenter (Altitudes)



Lots of Similar Triangles

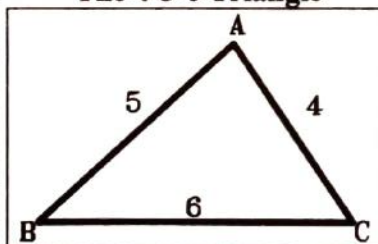
Circumcenter (⊥ bisectors)



$$K_{ABC} = \frac{ABC}{4R}$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$$

The 4-5-6 Triangle

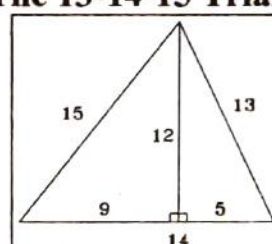


$$\angle BAC = 2\angle ABC$$

$$R = \frac{32\sqrt{7}}{7} \quad r = \frac{\sqrt{7}}{2}$$

$$K = \frac{15\sqrt{7}}{4}$$

The 13-14-15 Triangle

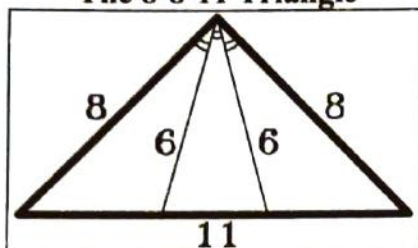


$$K = 84$$

$$R = \frac{65}{8}$$

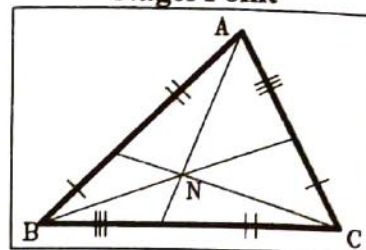
$$r = 4$$

The 8-8-11 Triangle



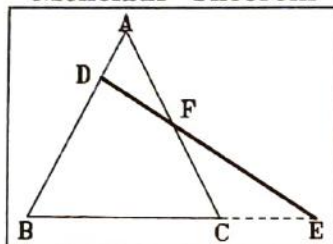
The trisectors of the biggest angle has length 6

Nagel Point



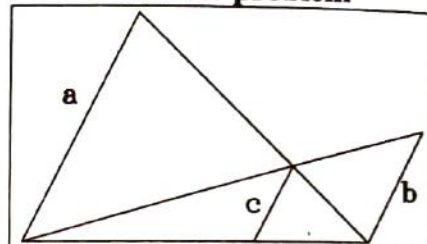
Joints semi-Perimeter points to Vertices

Menelaus' Theorem



$$\frac{AD}{DB} * \frac{BE}{EC} * \frac{CF}{FA} = 1$$

Two Pole problem



If $a \parallel b \parallel c$, then

$$\frac{1}{c} = \frac{1}{b} + \frac{1}{a}$$

In an equilateral triangle, the sum of the distances from any interior point to the three sides is always equal to the altitude of the triangle.

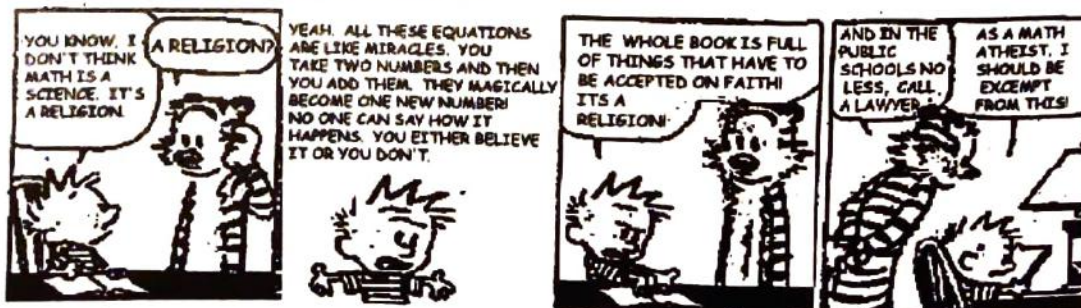
For triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$

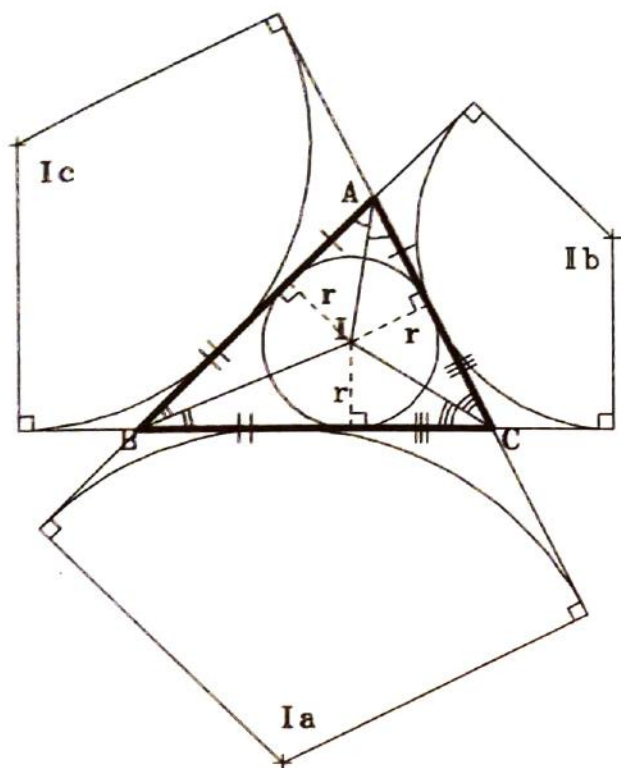
$$K = \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

Vertices of the centroid is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right)$$

CALVIN AND HOBBS By Bill Watterson





A Triangle and Its Circles

$\triangle ABC$ has sides a , b and c and angles A , B , and C .

The radius of the inscribed circle is r .

The radius of the circumscribed circle is R .
(see page 41)

The area of the triangle is K .

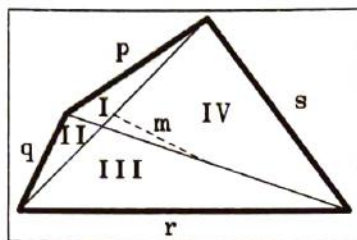
The semiperimeter of the triangle is S .

The altitude to sides a , b , c are h_a , h_b , h_c respectively.

The angle bisectors to side a , b , c are t_a , t_b , t_c respectively.

Area of Triangle	
Heron's Formula:	$K = \sqrt{S(S-a)(S-b)(S-c)}$
$K = \frac{1}{2}ab \sin C$	$K = \frac{a^2 \sin B \sin C}{2 \sin A}$
$K = rS$	$K = \frac{a \cdot b \cdot c}{4R}$
Assorted Identities	
$t_c = \frac{2a \cdot b \cdot \cos \frac{C}{2}}{a+b} = \frac{2\sqrt{a \cdot b \cdot S \cdot (S-c)}}{a+b}$	$r^2 = \frac{(S-a)(S-b)(S-c)}{S}$
$\frac{1}{r} = \frac{1}{h_a} + \frac{1}{h_b} + \frac{1}{h_c}$	
Law of Cosines	$c^2 = a^2 + b^2 - 2 \cdot a \cdot b \cdot \cos C$
Extended Law of Sines	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C} = 2R$
Law of Tangents	$\frac{a-b}{a+b} = \frac{\tan \frac{1}{2}(A-B)}{\tan \frac{1}{2}(A+B)}$
Second Law of Tangents (if $A + B + C = 180$)	$\tan A \cdot \tan B \cdot \tan C = \tan A + \tan B + \tan C$

Section H – Quadrilaterals

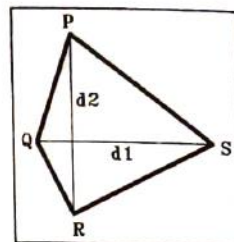


$$K_1 * K_3 = K_2 * K_4$$

$$K = \frac{1}{2} d_1 d_2 \sin \alpha$$

$$p^2 + q^2 + r^2 + s^2 = d_1^2 + d_2^2 + (2m)^2$$

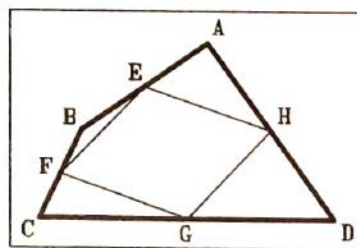
where m is connects the midpoint of the two diagonals



$$d_1 \perp d_2$$

$$K = \frac{1}{2} * d_1 * d_2$$

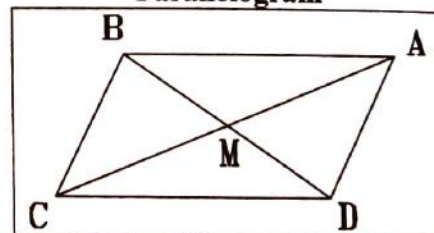
$$\overline{PQ}^2 + \overline{RS}^2 = \overline{PS}^2 + \overline{QR}^2$$



$$\text{If } \frac{\overline{AH}}{\overline{HD}} = \frac{\overline{DG}}{\overline{GC}} = \frac{\overline{CF}}{\overline{FB}} = \frac{\overline{BE}}{\overline{EA}} = n$$

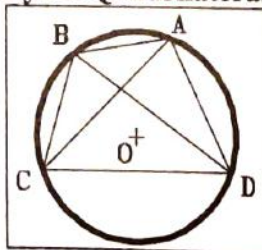
$$\text{then: } \frac{K_{EFGH}}{K_{ABCD}} = \frac{n^2 + 1}{(n + 1)^2}$$

Parallelogram



$$2(\overline{BC}^2 + \overline{BA}^2) = \overline{BD}^2 + \overline{AC}^2$$

Cyclic Quadrilateral



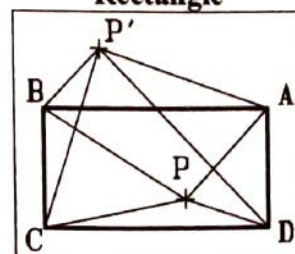
$$\angle A + \angle C = \angle B + \angle D = 180^\circ$$

$$K = \sqrt{(s-a)(s-b)(s-c)(s-d)}$$

Special case of Ptolemy:

$$\overline{BC} * \overline{AD} + \overline{AB} * \overline{CD} = \overline{BD} * \overline{AC}$$

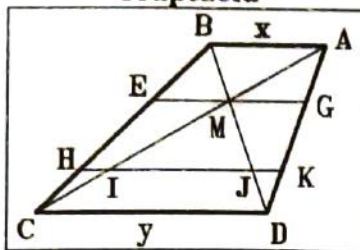
Rectangle



For any point P

$$\overline{PA}^2 + \overline{PC}^2 = \overline{PB}^2 + \overline{PD}^2$$

Trapezoid

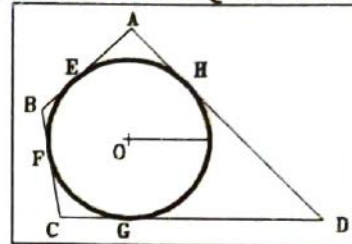


If $\overline{HK}$ is any parallel, then $HK = \frac{\overline{BH} * y + \overline{HC} * x}{\overline{BC}}$

If $\overline{HK}$ is a median, then $\overline{HK} = \frac{x+y}{2}$

$$\overline{EG} = \frac{2 * x * y}{x + y}$$

Circumscribed Quadrilateral



$$\overline{AB} + \overline{CD} = \overline{BC} + \overline{AD}$$

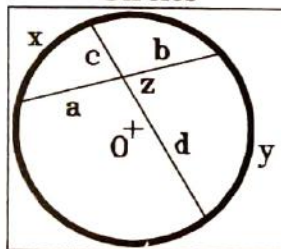
Quad_{ABCD} has incircle

$$r = \frac{K}{\overline{AB} + \overline{CD}} ; K = \frac{1}{2} r * P$$

If Quad_{ABCD} is both also inscribed, then K

$$= \sqrt{\overline{AB} * \overline{CD} * \overline{BC} * \overline{AD}}$$

Circles

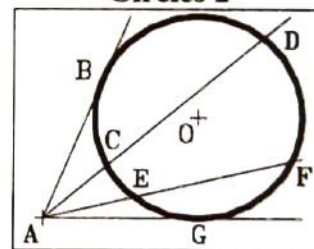


$$\angle c = \angle z = \frac{x + y}{2}$$

Power of the point

$$a * b = c * d$$

Circles 2



$$\overline{AB} = \overline{AG}$$

$$\angle DAF = \frac{\widehat{DF} - \widehat{CE}}{2}$$

$$\overline{AB}^2 = \overline{AD} * \overline{AC} = \overline{AF} * \overline{AE}$$

Ptolemy's Theorem

In any quadrilateral ABCD, $\overline{AC} * \overline{BD} \leq \overline{AB} * \overline{CD} + \overline{AD} * \overline{BC}$, with equality holding iff quadrilateral ABCD can be inscribed in a circle.

For a quadrilateral to be cyclic, it is necessary and sufficient condition that its opposite angles are supplementary.

In a circle of radius r, the length of a chord which subtends an inscribed angle θ is $2r \sin \theta$.

Section I – Trigonometry

Identities

- $\sin A = \frac{\text{opposite}}{\text{hypotenuse}}$
- $\cos A = \frac{\text{adjacent}}{\text{hypotenuse}}$
- $\tan A = \frac{\text{opposite}}{\text{adjacent}}$
- SOH-CAH-TOA
- $\sin^2 A + \cos^2 A = 1$
- $1 + \tan^2 A = \sec^2 A$
- $1 + \cot^2 A = \csc^2 A$
- $\tan A = \frac{\sin 2A}{1 + \cos 2A}$
- $\tan A = \frac{1 - \cos 2A}{\sin 2A}$

x	sin x	cos x	tan x
15	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$2-\sqrt{3}$
18	$\frac{\sqrt{5}-1}{4}$	$\frac{\sqrt{2(5+\sqrt{5})}}{4}$	$\frac{(\sqrt{5}-1)\sqrt{2}}{2\sqrt{5}+\sqrt{5}}$
30	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{3}}{3}$
36	$\frac{\sqrt{2(5-\sqrt{5})}}{4}$	$\frac{\sqrt{5}+1}{4}$	$\frac{\sqrt{2(5-\sqrt{5})}}{\sqrt{5}+1}$
45	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
75	$\frac{\sqrt{6}+\sqrt{2}}{4}$	$\frac{\sqrt{6}-\sqrt{2}}{4}$	$2+\sqrt{3}$

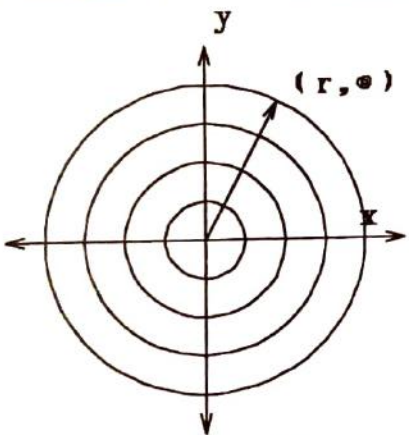
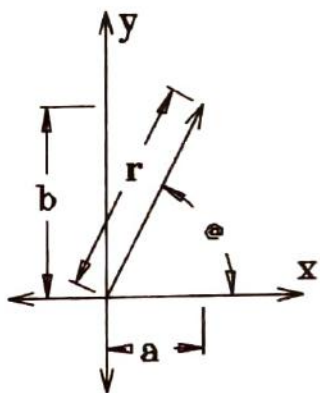
Odd-Even Functions		Sum & Difference Identities	
$\sin(-A) =$	$-\sin(A)$	$\sin(A \pm B) =$	$\sin A \cos B \pm \cos A \sin B$
$\cos(-A) =$	$\cos(A)$	$\cos(A \pm B) =$	$\cos A \cos B \mp \sin A \sin B$
$\tan(-A) =$	$-\tan(A)$	$\tan(A \pm B) =$	$\frac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$

Sum to Product		Product to Sum	
$\sin A \pm \sin B =$	$2 \sin \frac{A \pm B}{2} \cos \frac{A \mp B}{2}$	$\sin A \sin B =$	$\frac{1}{2} [\cos(A-B) - \cos(A+B)]$
$\cos A + \cos B =$	$2 \cos \frac{A+B}{2} \cos \frac{A-B}{2}$	$\cos A \cos B =$	$\frac{1}{2} [\cos(A-B) + \cos(A+B)]$
$\cos A - \cos B =$	$-2 \sin \frac{A+B}{2} \sin \frac{A-B}{2}$	$\sin A \cos B =$	$\frac{1}{2} [\sin(A-B) + \sin(A+B)]$
$\tan A \pm \tan B =$	$\frac{\sin(A \pm B)}{\cos A \cos B}$	$\tan A \tan B =$	$\frac{\cos(A-B) - \cos(A+B)}{\cos(A-B) + \cos(A+B)}$

Double Angle Identities		Triple Angle Identities	
$\sin(2A) =$	$2\sin(A) * \cos(A)$	$\sin(3A) =$	$3\sin A - 4\sin^3 A$
$\cos(2A) =$	$\cos^2 A - \sin^2 A$	$\cos(3A) =$	$4\cos^3 A - 3\cos A$
$\tan(2A) =$	$\frac{2 \tan A}{1 - \tan^2 A}$	$\tan(3A) =$	$\frac{\tan A(\tan^2 A - 3)}{3 \tan^2 A - 1}$

Half Angle Identities		Square Identities	
$\sin \frac{x}{2} =$	$\pm \sqrt{\frac{1 - \cos x}{2}}$	$\sin^2 x =$	$\frac{1}{2}(1 - \cos 2x)$
$\cos \frac{x}{2} =$	$\pm \sqrt{\frac{1 + \cos x}{2}}$	$\cos^2 x =$	$\frac{1}{2}(1 + \cos 2x)$

Section J – Polar and DeMoivre's Theorem

	
$x^2 + y^2 = r^2$ $\tan \theta = \frac{y}{x}$ $x = r * \cos \theta \quad y = r * \sin \theta$	$Z = a + bi = r * \text{cis} \theta$ (polar form of a complex number) The magnitude of Z, represented by $ a+bi = \sqrt{a^2 + b^2}$

Euler's Theorem	
	$e^{i\theta} = \cos \theta + i * \sin \theta = \text{cis } \theta$
$\text{cis}(\alpha + \beta) =$	$\text{cis}(\alpha)\text{cis}(\beta)$
$\text{cis}(\alpha - \beta) =$	$\frac{\text{cis } \alpha}{\text{cis } \beta}$
DeMoivre's Theorem	
$(a + bi)^n =$	$(r * \text{cis } \theta)^n = r^n * \text{cis}(n * \theta)$

Math Survey
would like to give
special thanks to the
Student Union
for its support of this
year's publication.



Almost Done...